Highlights

Standard validation overestimates cross-basin landslide susceptibility transfer: a rigorously evaluated few-shot benchmark across 14 Chilean watersheds

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- Five standard evaluation conventions inflate cross-basin landslide transfer.
- Selecting features with target labels doubled the few-shot gain (+2.4 to +4.7 pp).
- Leakage-free, buffered spatial CV: meta-learning +2–4 pp at K=1 only, 3/4 basins (Holm).
- Basin-balanced pooled pre-training reproduces the meta-learning gain; episodes add 0.
- Pooled logistic regression matches or beats every neural method at K=1 in all basins.

Standard validation overestimates cross-basin landslide susceptibility transfer: a rigorously evaluated few-shot benchmark across 14 Chilean watersheds

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Abstract

Transfer of landslide susceptibility models to newly studied basins is routinely assessed with conventions that overstate it. We quantify five of them on one benchmark—14 Chilean watersheds, ten source basins spanning the Atacama-to-Patagonia gradient and four semi-arid targets in north-central Chile (27–31°S)—using few-shot meta-learning as the paradigm under scrutiny and a 30 m MERIT-DEM feature stack. Random cross-validation inflates F_1 by up to 21 pp in heterogeneous basins; fixed-threshold evaluation hides that precision at 5% prevalence is 0.05–0.09; pooling correlated episodes narrows confidence intervals 3.3×; selecting features with target-basin labels doubles the apparent few-shot advantage (from +2.4 to +4.7 pp at $K = 1$, identical seeds); and comparing episodic meta-training against a size-weighted pooled baseline credits the algorithm with an effect of domain weighting. Under the corrected protocol—source-only feature selection, spatially disjoint and buffered validation, partition-level inference, five seeds—a meta-learned initialisation exceeds training from scratch at one labelled event per class by +4.3 pp F_1 jointly across the targets ($p < 0.001$; significant in three of four basins after Holm correction), by +1.6 pp at five events and by nothing at ten; an independent replication gives +2.2 pp. Pooled pre-training reproduces the advantage once source basins are weighted equally (−0.04 pp between episodic and balanced pooled training), a pooled-source logistic regression matches or exceeds every neural method at $K = 1$, and adversarial domain adaptation offers no few-shot gain. Benchmark, data and protocol are released.

Keywords: meta-learning, landslide susceptibility, transfer learning, remote sensing, spatial cross-validation, few-shot learning, Andes

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1. Introduction

Remote sensing has reshaped landslide susceptibility mapping by delivering globally consistent inputs: Sentinel-1 C-band SAR products for surface-deformation and amplitude features (Poggi et al., 2026), Sentinel-2 multi-spectral composites for surface cover and vegetation indices, MERIT and Copernicus 30 m DEMs for topographic and hydrologic derivatives, and WorldClim 2.1 bioclimatic surfaces for environmental context. These data, harmonized to a common grid, feed the dominant paradigm of fitting a machine-learning classifier to a per-basin inventory of labeled events—a workflow that produces strong results when local training data are abundant (Reichenbach et al., 2018; Merghadi et al., 2020; Ma et al., 2025), and that sits within the broader trajectory of deep-learning adoption across photogrammetric and remote-sensing tasks documented by Heipke and Rottensteiner (2020). However, the practitioner facing a *newly-studied* basin confronts a data-scarcity bottleneck: digitizing a usable inventory typically requires expert geomorphological mapping over hundreds of events, an effort that can take months and is rarely funded for individual watersheds. As a result, basins with sparse inventories ($N < 50$ mass-movement records) cannot benefit from data-hungry deep learning approaches, while transfer of models trained on data-rich neighboring basins suffers from systematic shifts in geology, climate, and surface cover (Zhu et al., 2026).

Whether a model transfers across basins is, however, a question that can only be answered as well as it is measured, and the conventions under which cross-basin transfer is usually measured are permissive. Spatially explicit cross-validation, essential for unbiased evaluation on autocorrelated Earth-observation data (Brenning, 2005; Roberts et al., 2017; Ploton et al., 2020), remains under-applied in landslide remote sensing; benchmarks are routinely reported on balanced test sets at a fixed threshold, with episodes or folds pooled as though they were independent, with negatives drawn far from the positives, and with features selected on the pooled data of all study areas, including the ones later used to test transfer. Each of these conventions biases the reported skill upward. Their combined effect on *cross-basin* transfer has not, to our knowledge, been quantified, and a transfer result that survives only under permissive validation cannot support the operational claim it is usually made to carry. This paper asks how much the standard conventions overstate cross-basin transfer, and what survives when they are removed.

We use few-shot transfer with meta-learning as the test case because it is the paradigm that promises the most from the least data. Two strategies have been used to mitigate data scarcity in new basins. **Transfer learning** by fine-tuning a model pre-trained on data-rich basins (Ghorbanzadeh et al., 2021) reduces label requirements but still needs tens to hundreds of target samples. **Domain adaptation** approaches such as DANN (Ganin and Lempitsky, 2015) learn domain-invariant representations from labeled source data and unlabeled target data, with mixed success in remote sensing (Tuia et al., 2016; Huang et al., 2026); source-free adaptation for cross-region Sentinel time-series mapping documents how shifts in $p(x)$ and $p(y \mid x)$ dominate cross-site transfer (Mohammadi et al., 2024). **Meta-learning** (Finn et al., 2017; Nichol et al., 2018) trains a model to *adapt rapidly to new tasks from a few examples*. It has shown promise in remote sensing for land-cover classification

(Rußwurm et al., 2020), satellite image classification (Tseng et al., 2023) and cross-scenario trait estimation (Fu et al., 2025), and it has reached landslide susceptibility in two forms: Chen et al. (2022) meta-learned an intermediate representation for few-shot susceptibility prediction across spatial blocks of two large study areas, and Ma et al. (2025) applied temporal meta-learning to dynamic susceptibility at a single site in Hong Kong. Neither evaluated transfer between hydrologically distinct watersheds, under spatially disjoint validation, against domain-adaptation and classical baselines, nor quantified how the evaluation protocol itself shapes the apparent advantage. This is the gap we address.

Using landslide susceptibility across 14 Chilean watersheds as the test bed—ten source basins spanning a $\sim$32° hydro-climatic gradient and four semi-arid target basins of north-central Chile—and first-order meta-learning as the transfer paradigm under scrutiny, we measure how much each evaluation convention inflates apparent transfer, and re-examine the methods under the stricter protocol. Our contributions are:

1. We **quantify how much five standard evaluation conventions inflate apparent cross-basin transfer**, each isolated as a controlled contrast on identical data: random versus spatially disjoint validation (up to +21 pp F_1 in the most heterogeneous basin, and no inflation in another); balanced test sets at a fixed threshold versus operational prevalence (precision falls to 0.05–0.09); pooling correlated episodes versus a partition-level bootstrap (confidence intervals 3.3× too narrow); selecting features with the labels of the target basins versus with source labels only (the $K = 1$ meta-learning advantage doubles, from +2.4 to +4.7 pp, under identical seeds); and a size-weighted pooled baseline versus a basin-balanced one (the apparent advantage of episodic meta-training over pooled pre-training disappears). We further verify that neither the adaptation budget, the negative-sampling rule, the number of spatial folds, nor exclusion buffers of up to 20 km around the held-out area manufacture the reported advantage. Together these define an evaluation protocol for cross-basin transfer that we release with the benchmark.
2. We construct a unified, balanced dataset spanning **14 Chilean basins** (10 source, 4 target) with a consistent 58-feature cross-basin schema—30 m MERIT-DEM derivatives (terrain, hydrology, texture, focal statistics), WorldClim bioclimatic variables, and SERNAGEOMIN geology categoricals—of which source-only feature selection retains nine morphometric descriptors. Inventory labels span four published catalogs and one ad-hoc digitization (4,591 positive and 4,591 negative pixels after balanced sampling, documented basin by basin).
3. We benchmark **five methods**—Independent (no source), Fine-tune (pooled pre-training), Reptile, FOMAML, DANN—plus a basin-balanced pooled baseline, a pooled-then-episodic baseline, three further meta-learning and adaptation baselines (ProtoNet, Meta-Baseline, CDAN) and **four classical machine-learning baselines** (logistic regression, random forest, XGBoost, CatBoost), across K-shot evaluation ($K \in \{1, 5, 10, 20\}$) on four target basins, with five seeds, hierarchical-bootstrap intervals, a pre-specified joint hypothesis test and Holm-corrected secondary tests (81,600 K-shot model fits in the core benchmark and 215,700 including the controls).

4. Our findings under the strict protocol: (i) a meta-learned initialisation (Reptile) exceeds training from scratch at $K = 1$ by +4.3 pp F_1 jointly across the four targets ($p < 0.001$; +2.6 to +7.4 pp per basin, significant in three of four after Holm correction), by +1.6 pp at $K = 5$, and by nothing at $K \geq 10$; an independent replication of the protocol with different initialisations gives +2.2 pp at $K = 1$ and one-hot encoding the single categorical input halves it, so the advantage is best described as 2–4 pp; (ii) this advantage is reproduced by pooled pre-training once source basins are weighted equally—episodic training and basin-balanced pooled training are indistinguishable (−0.04 pp)—so what transfers is a basin-balanced source representation, not the episodic optimisation; (iii) a pooled-source logistic regression matches or exceeds every neural method at $K = 1$ in all four targets; (iv) domain-adversarial alignment does not help at $K = 1$ and is weakly positive at $K = 5$; (v) the gains in ROC-AUC (+0.05 to +0.14) and in prevalence-weighted average precision survive re-weighting to a 5% landslide prevalence, where absolute precision collapses to 0.05–0.09 for every method.

The remainder of this paper is organized as follows. Section 2 reviews related work in landslide ML, transfer learning, meta-learning and validation bias. Section 3 describes the study area and data. Section 4 details the methodology. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Landslide susceptibility modeling

Statistical and machine-learning approaches to landslide susceptibility have been the subject of extensive review (Reichenbach et al., 2018; Merghadi et al., 2020). The field has progressively moved from bivariate and weights-of-evidence approaches to logistic regression, random forests, gradient-boosted trees (XGBoost, CatBoost), and more recently to deep neural networks. Brenning (2005) established spatial cross-validation as a methodological best practice for landslide modeling, demonstrating that random hold-out validation systematically overestimates generalization error due to spatial autocorrelation between nearby training and test pixels. Object-oriented approaches (Stumpf and Kerle, 2011) and patch-based convolutional models have refined the spatial feature representation, but the fundamental requirement of hundreds to thousands of labeled events per study area remains a binding constraint. Recent benchmarks comparing classical and ensemble ML algorithms (Merghadi et al., 2020) report F_1 scores in the 0.75–0.90 range when $N > 200$, but performance degrades sharply for small inventories. Most of the field operates within a single basin or contiguous region; cross-basin transfer is studied less frequently and is typically framed as out-of-sample testing rather than as a problem requiring methodological adaptation. Inventories augmented with InSAR-detected deformation (Chen et al., 2023) and multi-decadal dynamic susceptibility mapping (Ma et al., 2025) illustrate how richer labels and temporal structure are entering the field, which makes the question of how transfer is evaluated more, not less, pressing.

2.2. Transfer learning in remote sensing

Transfer learning by fine-tuning a model pre-trained on data-rich source domains is the de facto strategy for label-efficient remote sensing (Tuia et al., 2016). In landslide applications, Ghorbanzadeh et al. (2021) demonstrated that ImageNet-pretrained convolutional networks fine-tuned on a few hundred local samples outperform train-from-scratch baselines, confirming that representations learned on natural images contain transferable low-level features. However, fine-tuning a deep network with $K < 50$ target labels typically results in catastrophic over-fitting unless aggressive regularization or early stopping is applied. Furthermore, the "concat-source" variant—pre-training on a union of source basins, which is the baseline we adopt—implicitly assumes that all source domains are equally relevant to the target, a hypothesis that is rarely tested explicitly and that fails when the source pool spans heterogeneous environmental conditions; it also weights each source domain by its sample count, so that a single large inventory can dominate the pooled representation, a confound we isolate in Section 5.4. Recent work has begun to formalize cross-region transferability through spatiotemporal deep-learning frameworks that integrate environmental context (Luo et al., 2026), transferable landslide segmentation via universal adapters (Wei et al., 2024), cross-resolution landslide mapping (Ng et al., 2025), and adversarial domain adaptation (Zhang et al., 2025); semi-supervised event detection from multi-temporal optical imagery offers a complementary path for sparse-inventory settings (Deijns et al., 2024). However, these approaches target dense segmentation from imagery, and landslide-specific cross-basin transfer on point inventories with rigorous spatial validation remains an open question.

2.3. Domain adaptation

Domain adaptation (DA) seeks to learn classifiers that generalize across distribution shifts by leveraging both labeled source data and unlabeled target data. The Domain-Adversarial Neural Network (DANN) of Ganin and Lempitsky (2015) is the canonical method: a feature extractor is trained simultaneously to (i) discriminate the class label and (ii) confuse a domain classifier through a Gradient Reversal Layer, encouraging the learned representation to be domain-invariant. Variants based on Maximum Mean Discrepancy (Long et al., 2015) and Wasserstein distance have followed. In remote sensing, DA has been applied to multi-source land-cover classification, image segmentation, and scene recognition, with mixed success (Tuia et al., 2016): performance gains are typically substantial when source domains are abundant and labeled target data are completely absent, but degrade when only a handful of source domains are available or when small numbers of labeled target samples are accessible. The latter regime is precisely our setting, and our results below show that adversarial alignment yields no advantage at $K = 1$ and a small one at $K = 5$ when the source pool has ten domains.

2.4. Meta-learning

Meta-learning frames the data-scarcity problem as one of *learning to learn*: the model is trained on a distribution of related tasks so that adaptation to a new task from few examples requires only a small parameter update. Model-Agnostic Meta-Learning (MAML;

Finn et al. 2017) is the canonical second-order method, computing gradients through inner adaptation steps, but it is computationally expensive and unstable in practice. First-order approximations—FOMAML, which drops the second-order term, and Reptile (Nichol et al., 2018), which replaces query-loss-driven updates with averaged inner-step displacements—are more practical. Theoretical work (Raghu et al., 2020) suggests that MAML's success can be largely attributed to feature reuse rather than rapid weight adaptation, which we revisit in our adaptation-curve analysis (Section 5).

In remote sensing, meta-learning has been applied to few-shot land-cover classification (Rußwurm et al., 2020), achieving 70–90% accuracy with $K = 5$ samples per class on EuroSAT and Sen12MS benchmarks, and to satellite image classification more broadly (Tseng et al., 2023). For landslide susceptibility, Chen et al. (2022) segmented two large study areas into blocks with distinct landslide-inducing environments and meta-learned an intermediate representation from which block-specific models are adapted with very few samples, demonstrating few-shot adaptation to unseen blocks; Ma et al. (2025) demonstrated temporal meta-learning for dynamic susceptibility mapping at a single site (Hong Kong), decomposing yearly subtasks and achieving fast adaptation in 5 samples and 5 inner steps. Our work differs from both in the unit of transfer and in the evaluation: the tasks are whole watersheds that differ in climate, lithology and inventory source rather than blocks or years of one area; the targets are evaluated under spatially disjoint, buffered and leakage-controlled validation with partition-level inference; and the meta-learners are compared against domain-adaptation, basin-balanced pooled and classical baselines on identical partitions and support draws. The first two studies established that meta-learning can adapt susceptibility models from few samples; the present one asks how much of that advantage survives rigorous evaluation when the new task is a different watershed.

2.5. Validation bias in spatial prediction

The upward bias of random cross-validation on spatially autocorrelated data is well established in ecology and soil science (Roberts et al., 2017; Ploton et al., 2020). Meyer and Pebesma (2021) formalised the distinction between interpolation within the training extent and extrapolation beyond it, and showed that the choice of cross-validation design determines which of the two is being measured. Wadoux et al. (2021) argue, conversely, that spatial cross-validation is not the right tool for estimating map accuracy over a defined area, for which a probability sample is needed; we therefore use spatial cross-validation to compare methods under matched conditions, not to certify the accuracy of any map. Beyond the fold design, several other conventions inflate reported skill: evaluation on balanced test sets at a fixed threshold ignores the extreme class imbalance of operational mapping; pooling correlated repetitions (episodes within a fold, seeds within a region) understates uncertainty; negatives drawn far from positives create a steep-versus-flat problem easier than the real one (Abdelkader et al., 2025); and feature selection on the pooled data of all study areas lets information from the test areas enter the model before evaluation. Each of these has been discussed in isolation; their joint effect on a cross-basin transfer benchmark, and the question of which transfer claims survive their removal, is the subject of this paper.

3. Study area and data

3.1. Basins

We study 14 watersheds distributed along the length of Chile (Figure 1), from the hyper-arid Atacama Desert (∼21°S) to Patagonia (∼53°S), selected to span an extreme hydro-climatic gradient (basin-mean annual precipitation ∼1 to ∼1,900 mm/yr, with individual pixels in the wettest basin exceeding 2,500 mm/yr) and to have heterogeneous landslide-mapping coverage. **Ten source basins** are used for meta-training: Chañaral / Río Salado (BNA 032, hyper-arid; 800 landslide and 910 non-landslide points from the supplementary data of Parra (2025), mapped after the March 2015 Atacama floods), Costeras Q. Negra–Pan de Azúcar / Taltal (BNA 029, hyper-arid coastal; 80 mass-movement polygons + 300 negatives from Parra (2025)), Costeras Tilviche-Loa (BNA 018; 127 events), Costeras Loa-Caracoles (BNA 022; 50 events), Río Maipo (BNA 057; 349 events), Río Rapel (BNA 060; 392 events), Río Maule (BNA 073, semi-humid; the 203 of the 1,227 events in the 2010 M_w 8.8 earthquake-induced inventory of Serey et al. (2019) that fall within the basin polygon, retaining all four mapped typologies—disrupted slides, flows, lateral spreads and coherent slides), Río Choapa (BNA 047, semi-arid; 62 events from the SERNAGEOMIN national *catastro de remociones en masa*), Costeras Bueno-Puelo (BNA 104; 47 events), and Costeras Magallanes (BNA 125; 413 events). **Four target basins** are used for K-shot evaluation: Río Copiapó (BNA 034; 21 catastro events, 19 after snapping to the grid), Río Huasco (BNA 038; 296 events, 278 pixels), Río Elqui (BNA 043; 141 events, 132 pixels), and Río Limarí (BNA 045; 42 events, 35 pixels).

The four targets are the semi-arid basins of north-central Chile (27–31°S) that the funding programme designates as priority basins for susceptibility mapping, and the only basins in the national catastro with inventories independent of the source data. The source pool, not the target set, spans the Atacama-to-Patagonia gradient: the source basins supply the climatic and lithological diversity from which a transferable initialisation is learned, whereas transfer is *evaluated* only on semi-arid targets. Transfer to humid or Patagonian basins is therefore not tested here, and the conclusions are stated for the evaluated targets. Table A.7 (Appendix Appendix A) summarises, for every basin, the inventory source and geometry, the number of mapped events, the negative-sampling rule and the dominant trigger reported by the inventory. Two properties of this dataset bound the generality of the results and are returned to in Section 6.6: the inventories mix rainfall- and earthquake-triggered movements of several types, and one polygon inventory (Taltal) contributes 49% of the source positives.

3.2. Remote-sensing and geospatial data sources

All conditioning factors are derived from open Earth-observation products harmonized to a common 30 m raster grid in UTM zone 19S. Table 1 summarizes the five primary sources, their spatial and temporal coverage, and the feature families derived from each.

The satellite-derived MERIT DEM and its terrain, hydrology and texture derivatives supply the bulk of the stack. MERIT DEM is preferred over individual SRTM tiles because its hydrologic corrections (pit filling, stream burning) yield more stable flow-routing

Table 1: Remote-sensing and geospatial inputs used to compute the per-basin feature stack. All products are public and openly accessible. Sentinel-2 optical composites were computed for two source basins only and are excluded from the common cross-basin schema (Section 3.3); the operative inputs are the satellite-derived MERIT DEM, its terrain/hydrology/texture derivatives, WorldClim climate and SERNAGEOMIN geology. After source-only feature selection (Section 3.5) the nine features used by all models are DEM derivatives.

Product	**Sensor / Source**	**Native res.**	**Period**	**Features derived**
MERIT DEM (filled, hydrology-corrected)	SRTM + AW3D + ICESat fusion	3 arc-sec (∼90 m)	static	28 terrain features (slope, aspect, MRVBF, openness, geomorphons, etc.); 10 hydrology features (flow accumulation, HAND, TWI, drainage density, sediment connectivity); 6 GLCM texture; 6 focal statistics
WorldClim 2.1	Multi-station interpolated climatology	2.5 arc-min (∼5 km)	1970–2000 baseline	5 bioclimatic variables (bio_01 mean temp; bio_12 annual precip; bio_13/14 wettest/driest month precip; bio_15 seasonality)
SERNAGEOMIN national geological map	National Geological Survey of Chile	1:1,000,000 vector	static	3 categorical features (lithology, rock type, geological age)
BNA basin polygons + landslide inventories	DGA + SERNAGEOMIN *catastro de remociones* + Serey et al. (2019) + Parra (2025)	vector	1922–2023 mixed	Basin extent for sampling; positive labels (4,591 pixels) and negatives (4,591 pixels)

derivatives essential for HAND and sediment connectivity. WorldClim 2.1 captures the 1970–2000 climatological baseline; while static climate features cannot represent extreme events that trigger individual landslides, they parameterize the long-term hydro-climatic regime that conditions susceptibility (Ma et al., 2025). All raster inputs are resampled bilinearly (continuous variables) or by nearest neighbor (categorical), then stacked into a per-pixel feature vector consumed by the classifier. We acknowledge that this introduces an effective-resolution mismatch: MERIT-derived terrain and texture features inherit the ∼90 m source resolution despite being sampled on the 30 m grid, and WorldClim bioclimatics are nearly piecewise-constant within each ∼5 km native cell. Topographic and texture derivatives can thus be expected to be smoother than a native 30 m DEM (e.g., Copernicus GLO-30) would yield, and the climate-distance analysis of Section 5.15 is interpreted at basin level rather than per pixel. We return to this limitation in Section 6.6. Intermediate per-basin products can be regenerated from public sources using the released pipeline (see the Data and code

availability statement at the end of the manuscript) to support reproducibility.

3.3. Conditioning factors

We compute a common cross-basin stack of **58 features** per basin at 30 m grid alignment, organized in six categories. To guarantee an identical feature schema across all 14 basins, we retain only features available in every basin (Section 3.4); Sentinel-2 spectral composites, computed for two coastal source basins (Taltal, Choapa) but not the remaining twelve, are therefore excluded by this intersection and do not enter the benchmark. The 58 retained features are:

- **Terrain (28):** slope, aspect, eastness, northness, hillshade, sky-view factor, openness (positive, negative), geomorphons, landform, Multi-resolution Index of Valley Bottom Flatness (MRVBF), Multi-resolution Index of Ridge-Top Flatness (MRRTF), vector ruggedness measure, deviation from mean elevation, convergence, curvature, curvedness, shape index, surface area ratio, three TPI radii plus TPI, valley depth, relative slope position, LS-factor, TRI.
- **Hydrology (10):** filled DEM, flow accumulation (D8, MFD), flow direction (D8, D∞), HAND, stream network, TWI, drainage density, sediment connectivity (Borselli et al., 2008).
- **Texture GLCM (6):** contrast, correlation, dissimilarity, energy, entropy, homogeneity, computed on the DEM rather than on optical imagery so that the texture descriptors capture topographic roughness (a physical correlate of slope instability) and remain available in every basin, independent of cloud-free optical coverage.
- **Focal statistics (6):** DEM std at radii 1, 4, 10; DEM range $r = 4$; slope std $r = 4$; slope mean $r = 4$.
- **Climate (5):** WorldClim 2.1 bioclimatics—bio_01 (annual mean temperature), bio_12 (annual precipitation), bio_13/14 (precipitation wettest/driest month), bio_15 (precipitation seasonality).
- **Geology (3):** SERNAGEOMIN 1:1,000,000 lithology class, rock type, geological age (categorical, encoded).

All factors are computed using SurtGIS, a Rust-based geospatial toolkit (Parra, 2026), with reprojection to UTM zone 19S and bilinear/categorical resampling as appropriate.

3.4. Sampling protocol

We adopt a **hybrid balanced sampling** strategy. Each pixel of the 30 m DEM grid is the unit of analysis. For each basin:

1. **Positives:** For point inventories, snap to nearest pixel (coincident points collapse to one pixel, which is why pixel counts are slightly below event counts). For polygon inventories (Taltal), rasterize with `all_touched=True` and randomly sub-sample at most 30 pixels per polygon to limit spatial autocorrelation.
2. **Negatives:** Only two basins provide mapped non-landslide points: Chañaral (910 points) and Taltal (300 points). There, the mapped points are expanded by a 60 m buffer to cover surrounding pixels and, in Taltal, completed by random sampling to reach the positive count. In the remaining twelve basins, negatives are generated by random sampling within the basin polygon, excluding a 500 m buffer around positives to avoid spatial leakage. The rule applied to each basin is listed in Table A.7; the sensitivity of the results to the negative-sampling rule is tested in Sections 5.1 and 6.6.
3. **Balancing:** Negatives are sub-sampled or augmented to match positive count exactly (1:1).

This procedure yields 9,182 samples across the 14 basins: 4,591 positive and 4,591 negative pixels, balanced 1:1 within every basin (per-basin counts in Table A.7). The ten source basins contribute 4,127 positives, of which Taltal's rasterised polygons supply 2,003 (49%) and Chañaral 562 (14%); the four target basins contribute 464 positives (Copiapó 19, Limarí 35, Elqui 132, Huasco 278). Target basins are used exclusively for K-shot evaluation: no target pixel or label enters meta-training, pre-training, feature selection or the estimation of any preprocessing parameter that uses labels.

Preprocessing. Continuous features are standardised per basin by subtracting the mean and dividing by the standard deviation of all sampled pixels of that basin, without using labels. For a target basin these statistics are computed over its sampled pixels, including those that later serve as queries; this is an unsupervised, domain-wise normalisation equivalent to standardising over the basin raster, which is available operationally before any inventory exists, and it uses no label information. Missing values (rare, at raster edges) are imputed with the per-basin feature median. One of the selected features, landform (11 classes), is categorical in nature; it was entered as integer class codes, a choice we test by one-hot encoding in Section 5.13, together with the alternative of estimating the standardisation statistics from the source pool or from the $2K$ support pixels only.

3.5. Feature selection via correlation-based feature selection (CFS)

We apply **correlation-based feature selection** (Hall, 1999) using symmetric uncertainty (information-theoretic, robust to mixed data types). Features are quantile-discretized into 10 bins. Forward greedy search adds the feature that maximizes the subset score:

$$M_S = \frac{k \cdot \overline{r_{cf}}}{\sqrt{k + k(k-1) \cdot \overline{r_{ff}}}}, \tag{1}$$

where k is subset size, $\overline{r_{cf}}$ the mean feature-class correlation, and $\overline{r_{ff}}$ the mean inter-feature correlation. The procedure terminates when no feature addition improves the score.

Feature selection uses the labels of the **ten source basins only** (8,254 samples). Run on the source pool, CFS selects **9 features**: 5 terrain (landform, openness positive, MRVBF, MRRTF, curvedness), 2 focal statistics (DEM std $r = 10$, slope std $r = 4$) and 2 texture GLCM (entropy, correlation). No hydrology, geology or climate feature is selected: the discriminative signal shared across the Chilean source inventories is entirely morphometric. In the originally submitted version of this work, CFS had been run on all 14 basins, so that the labels of the four target basins (10% of the samples) contributed to the choice of features; that run selected two additional features, geomorphons and annual precipitation (bio_12), both of which carry within-target signal (Table A.9). Selecting features with target labels is a form of leakage that precedes K-shot evaluation, and we quantify its effect on the reported transfer as one of the evaluation conventions studied in Section 5.3. All results in this paper use the source-only selection.

4. Methods

4.1. Backbone model

All methods share a 3-layer MLP backbone: 9 input features → 64 hidden → 32 hidden → 1 output, with ReLU activations, 10% dropout per hidden layer, and binary cross-entropy with logits as the training loss. We deliberately use a small tabular network rather than a CNN or Transformer because (i) our feature representation is point-wise and scale-invariant, (ii) our pixel-level samples lack spatial neighborhoods at this stage of the pipeline, and (iii) it ensures interpretability of feature importance. Section 6 discusses the natural extension to U-Net architectures for spatially-structured prediction.

4.2. Methods compared

Independent. Train MLP from scratch on K target labels (K positives + K negatives), 30 Adam steps at learning rate 10^{-2}. No source-basin information is used.

Fine-tune (concat-source baseline). Pretrain MLP on the concatenation of all 10 source basins (300 full-batch Adam steps at 10^{-3}), then fine-tune for 30 steps on K target samples. Because the concatenation is size-weighted, Taltal contributes 49% of the pre-training positives and the four smallest source basins together 7%.

Fine-tune (basin-balanced). Identical to Fine-tune except that the pre-training loss weights every sample by the inverse size of its basin, so that each source basin receives the same total weight as it does under episodic meta-training. This control separates the effect of *domain weighting* from the effect of *episodic optimisation* (Section 5.4).

Reptile. (Nichol et al., 2018). First-order meta-learning. For 300 outer steps:

1. Sample a source basin τ uniformly at random (every basin is equally likely regardless of its size).
2. Sample $K_{\text{meta}} = 10$ support points.
3. Adapt model with 5 inner SGD steps at 10^{-2}.

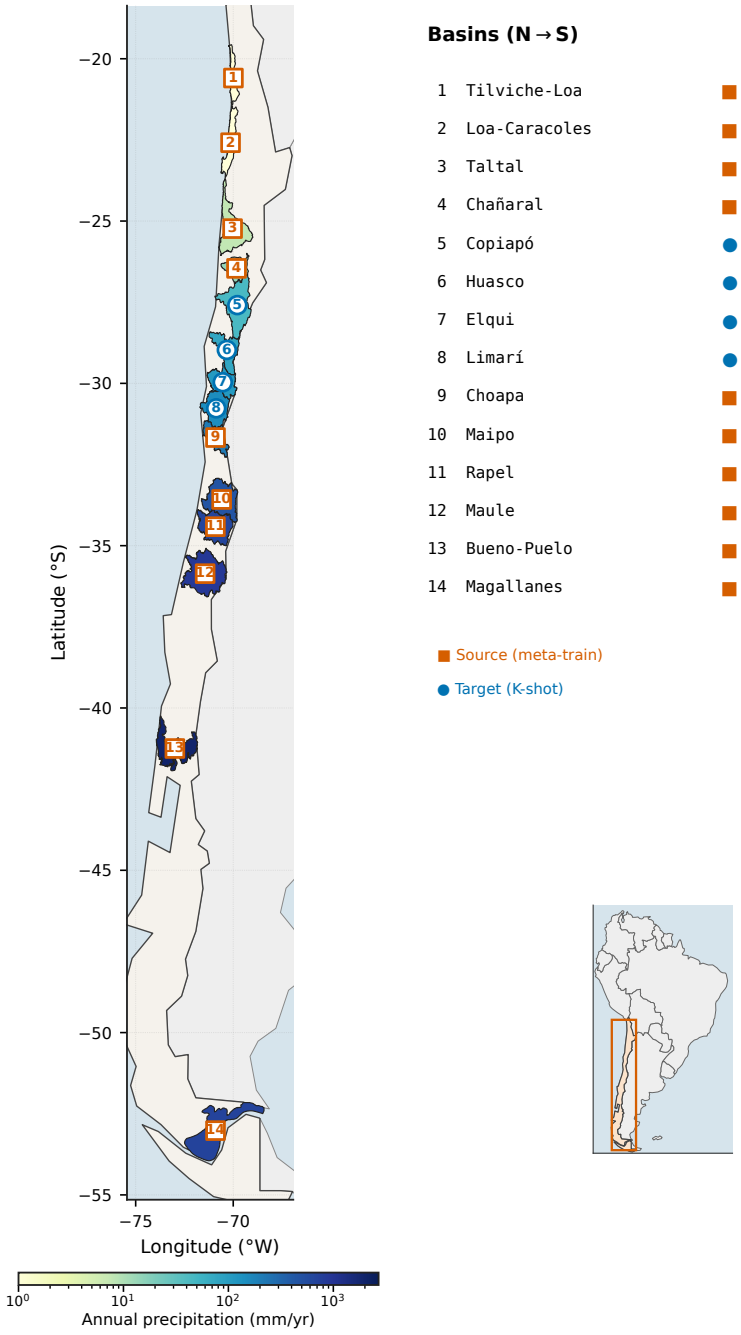


Figure 1: Fourteen Chilean watersheds spanning an extreme climate gradient—from the hyper-arid Atacama in the north (∼1 mm/yr) to the humid Patagonian Andes in the south (∼1,900 mm/yr basin mean), over ∼32° of latitude. Basins are numbered north-to-south; source basins (orange squares) are used for meta-training and target basins (blue circles) for K-shot evaluation. Polygons are color-coded by mean annual precipitation (WorldClim bio_12, log scale). Inset: location within South America.

4. Update meta-model: $\theta \leftarrow \theta + \epsilon(\theta_{\text{adapted}} - \theta)$ with $\epsilon = 0.1$.

At test time: load meta-model and adapt for 30 steps on K target samples (same protocol as Fine-tune).

Pretrain→Reptile. Reptile meta-training (300 outer steps, as above) initialised from the concat-source pre-trained weights instead of from a random initialisation. Together with Fine-tune, Fine-tune (balanced) and Reptile it completes a 2×2 design, {size-weighted, basin-balanced} × {pooled, episodic}, evaluated on identical partitions and support draws.

FOMAML (First-Order MAML). For 300 outer steps:

1. Sample task τ (a source basin, uniformly), support S, query Q.
2. Inner: 5 SGD steps on S, lr 10^{-2}.
3. Compute query loss on the *adapted* model.
4. Take query-loss gradient w.r.t. adapted parameters; copy this gradient to the meta-model.
5. Apply Adam outer-step at lr 10^{-3}.

DANN. (Ganin and Lempitsky, 2015). Feature extractor F: $9 \to 64 \to 32$ (shared). Class head C: $32 \to 1$. Domain head D: $32 \to 10$ (with Gradient Reversal Layer). Trained for 100 epochs at batch 64, lr 10^{-3}, with annealed $\lambda(p) = 2/(1 + e^{-10p}) - 1$. After training, $F + C$ are loaded into the MLP backbone for K-shot adaptation following the same protocol as Fine-tune.

ProtoNet. (Snell et al., 2017). Prototypical networks. A shared feature encoder (same $9 \to 64 \to 32$ architecture as the MLP backbone) is meta-trained episodically: for each source task, class prototypes are the mean support embeddings, and the prototypical loss is the cross-entropy of negative squared-Euclidean distances from query embeddings to prototypes (300 outer steps, Adam at 10^{-3}). At test time, the K target support points define prototypes and query pixels are classified by nearest prototype—no gradient adaptation is required.

Meta-Baseline. (Chen et al., 2021). The encoder is first pre-trained with standard supervised classification on the concatenated source pool, then frozen; at test time, K-shot cosine-similarity prototypes classify query pixels. This tests the claim that a strong pre-trained representation with a simple metric head can match episodic meta-learning.

CDAN. (Long et al., 2018). Conditional adversarial domain adaptation. The domain classifier is conditioned on the outer product of feature embeddings and class-probability predictions (rather than features alone as in DANN), which the authors argue yields better-aligned multimodal distributions. Training and K-shot adaptation follow the same protocol as DANN.

Spatial CNN (U-Net-style) backbone. As a methodological control on the point-wise MLP, we also train a small patch-classifier CNN on multi-channel image patches. Standard dense-segmentation benchmarks are not directly applicable to sparse point-inventory susceptibility, which has neither pixel-level masks nor bounding boxes; we therefore adopt a patch-classifier formulation that predicts per-point susceptibility from a local feature window. For each inventory point we extract a 32×32 window (covering $\sim 960 \times 960$ m at 30 m resolution) from the 25 gridded feature rasters (terrain + hydrology + texture + focal statistics) consistently available across all basins (this experiment does not use the CFS-selected features and is therefore unaffected by the feature-selection change of Section 3.5). The architecture is three Conv–ReLU blocks (32, 32, 64 channels) with two MaxPool layers, an adaptive global average pool, and a 32-unit FC head producing a binary logit ($\sim$ 74,000 parameters). Reptile, FOMAML, Fine-tune (concat-source pre-train) and Independent are applied to this backbone under the same spatial K-shot CV protocol on the largest target basin (Huasco), with five fewer episodes per fold (15 vs. 20) to keep compute bounded. Magallanes was excluded from the source pool for this experiment because its raster stack exceeds the worker memory budget. Results (Section 5.14) quantify the value of spatial context for sparse-inventory susceptibility transfer.

Classical ML baselines. For comparison, we evaluate four classical algorithms under identical K-shot spatial-CV protocol: **logistic regression** (sklearn, L_2 regularization, $C = 1.0$, 500 iterations); **random forest** (100 estimators, max depth 10); **XGBoost** (100 estimators, max depth 5, lr 0.1); **CatBoost** (100 iterations, depth 5, lr 0.1). Each algorithm is run in two modes: *independent* (trained only on K target samples) and *concat-source* (trained on full source pool + K target samples).

Pre-training budgets. All source-trained initialisations see the source pool for a comparable number of parameter updates: 300 full-batch Adam steps for the two pooled pre-trainings, 300 outer steps (each with 5 inner steps on 20 samples) for Reptile, FOMAML and ProtoNet, and 100 epochs at batch 64 for DANN and CDAN. Meta-training a tabular MLP on 8,254 source samples takes seconds of CPU time per seed for every method, so no method is favoured by compute.

Statistical protocol for extended baselines. ProtoNet, Meta-Baseline, and CDAN are evaluated under the same spatial-CV protocol as the primary methods with **five seeds** $\{42, 123, 7, 1729, 2718\}$, yielding 16,380 additional K-shot adaptations.

4.3. K-shot evaluation protocol

Throughout, K denotes the number of support examples *per class*: each K-shot episode uses K labelled positives and K labelled negatives ($2K$ support samples in total), so that $K = 1$ corresponds to a single verified mass-movement footprint plus one non-landslide pixel. For each (target basin, K, method, seed) combination, we run two evaluation protocols:

Random cross-validation. K-shot episodes with support and query sampled at random from the basin's pixel pool; spatial proximity is not enforced. The query is a balanced random draw of 10 positives and 10 negatives disjoint from the support. The realized benchmark comprises 40 episodes per (target, K, method, seed) with three seeds; Copiapó is restricted to $K \in \{1, 5\}$ by its 19-event inventory, yielding 8,400 adaptations in total. Random CV is reported only as the permissive foil against which the spatial protocol is compared.

Spatial cross-validation. The target basin is partitioned into 5 spatial clusters via K-means on (x, y) UTM coordinates. For each fold (held-out cluster), 20 episodes sample K positives + K negatives from the *training* clusters as support; the query is the *full held-out cluster*, which is neither balanced nor of fixed size: valid held-out clusters contain between 6 and 265 pixels and their positive share ranges from 3% to 86% with the local inventory density (Table A.8). This guarantees support and query are spatially disjoint, addressing spatial autocorrelation per Brenning (2005); Pohjankukka et al. (2017); Valavi et al. (2019). Fold × seed partitions whose held-out cluster lacks both classes, or whose training clusters cannot supply K positives and K negatives, yield no valid episodes and are skipped. With five seeds, the number of valid partitions is Elqui 25/25, Limarí 22/25, Copiapó 15/25 and Huasco 10/25; every exclusion is a single-class held-out cluster, and the excluded clusters are the low-relief, event-free parts of Huasco and Limarí (slope standard deviation 1.7 vs. 6.7 in Huasco; Table A.8, Figure B.8). The reported metrics therefore describe the event-bearing, mountainous parts of each basin, where F_1 is defined, rather than the basin as a whole. We acknowledge that K-means folds do not enforce a block size larger than the empirical variogram range of model residuals; per the framework of Meyer and Pebesma (2021), this implies that reported metrics describe interpolation within the basin's training extent rather than extrapolation. Section 5.6 measures the separation between query and support pixels directly and repeats the benchmark with exclusion buffers of 2–20 km around the held-out cluster.

Each episode:

1. Sample support S (K positives + K negatives).
2. For meta-trained methods (Reptile, FOMAML, DANN), load the corresponding meta-model; otherwise initialize from scratch.
3. Adapt with 30 Adam steps on S.
4. Evaluate F_1 and ROC-AUC on the query.

The random-number stream of every episode is seeded deterministically from (seed, fold, K, target), so that all methods evaluated in one script see the same support draws and every run is reproducible.

F_1 is computed at a fixed decision threshold of 0.5 on the query; because operational susceptibility mapping faces extreme class imbalance, this F_1 overstates the precision a practitioner would observe. We therefore also report the threshold- and prevalence-independent ROC-AUC in the main text, and a dedicated prevalence-robust analysis (precision and average precision re-weighted to realistic landslide prevalences; Section 5.7).

Replication structure and uncertainty. The replication structure is seed → spatial partition → fold → episode. Each seed draws a new K-means partition and a new meta-model, but partitions of different seeds can coincide: the adjusted Rand index between seeds is 1.00 in Huasco (identical partitions), 0.93 in Copiapó, 0.84 in Elqui and 0.70 in Limarí (Table A.8). Within one (seed, fold) partition, all 20 episodes evaluate on the same held-out cluster and draw supports from the same training pool, so episodes are not independent, and the effective number of spatial replicates lies between the number of folds (5) and the number of (seed, fold) partitions (10–25 per target), not at the nominal 200–500 runs. Every interval and test in this paper is therefore computed at the partition level: (i) all F_1 means in Table 2 and in the spatial-CV figures carry **two-stage hierarchical-bootstrap** 95% intervals, which resample (seed, fold) partitions with replacement and then episodes within each resampled partition (2,000 resamples); the pooled intervals that treat episodes as independent are a median 3.3× narrower and are reported only in the supplement; (ii) differences between methods are assessed with a **cluster-level paired test**, in which the per-partition mean-F_1 difference between a method and Independent, evaluated on the same held-out cluster, is one paired observation (Table 4); (iii) as a sensitivity analysis we also average seeds within each fold and treat the fold as the unit ($n = 5$ or fewer per target), the most conservative reading.

Pre-specified hypothesis and multiple comparisons. The primary hypothesis is a single joint statement: *Reptile exceeds Independent at $K = 1$ under spatial cross-validation, across the four target basins.* It is tested with a permutation test stratified by target: the sign of every partition-level paired difference is flipped at random within its target (under the null the method labels of a shared partition are exchangeable), the statistic is the mean over targets of the per-target mean difference, so that each basin has equal weight irrespective of its number of partitions, and p is computed over 20,000 permutations. Per-target comparisons for Reptile and FOMAML are secondary (eight tests) and are Holm-corrected; Fine-tune and DANN are reported descriptively. Comparisons at $K = 5$ and $K = 10$ use the same paired procedure.

The main spatial-CV K-shot benchmark (Table 2) nominally spans 4 targets × 4 K values × 5 methods × 5 seeds × 5 folds × 20 episodes = 40,000 adaptations; after excluding Copiapó $K = 20$ and infeasible fold × seed partitions, the realized benchmark comprises 27,300 K-shot adaptations. Adding the random-CV benchmark (8,400 adaptations), the classical-ML baselines (29,520 fits under both CV protocols and both train modes), and the extended meta-learning baselines (16,380 adaptations) yields 81,600 K-shot model fits; the controls introduced in this revision (domain-weighting design, buffered validation, leakage quantification) add a further 134,100, for 215,700 in total, complemented by the adaptation-curve experiment and the hyperparameter-sensitivity ablation grid.

5. Results

5.1. Spatial-CV K-shot benchmark

Table 2 and Figure 3 give the main result under spatially disjoint validation with the source-only feature set and five seeds. At $K = 1$, the meta-learned Reptile initialisation has the highest mean F_1 in all four target basins (Copiapó 0.594, Huasco 0.711, Elqui 0.470, Limarí 0.474, against Independent 0.519, 0.685, 0.434 and 0.436), with FOMAML second in three. The paired cluster-level differences to Independent (Table 4) are +7.4 pp in Copiapó, +2.6 in Huasco, +3.6 in Elqui and +3.8 in Limarí for Reptile, and +2.0 to +3.5 pp for FOMAML. Fine-tune, the size-weighted pooled pre-training, is within ±2 pp of Independent in every basin (+2.2, +0.1, +0.6, −0.6), and DANN is neutral or negative (+2.8, −2.5, −0.2, +1.3). The advantage decays with the support size: averaged over targets, Reptile exceeds Independent by +4.3 pp at $K = 1$, +1.6 pp at $K = 5$, −0.4 pp at $K = 10$ and +0.6 pp at $K = 20$, so the few-shot regime in which a source-learned initialisation pays off ends at about ten labelled events per class. Absolute discrimination remains modest in the two heterogeneous targets, Elqui and Limarí, where no method exceeds $F_1 \approx 0.50$ at $K = 1$ or 0.65 at $K = 20$.

The hierarchical intervals of Table 2 are wide—in Copiapó, whose 15 valid partitions each hold 7–12 pixels, they span ±0.1—and the intervals of Reptile and Independent overlap in every cell. This is expected: marginal intervals ignore the pairing on shared partitions and are not the appropriate significance criterion; the paired tests of the next section are.

5.2. Inference: joint test, per-basin tests and the replication unit

The pre-specified primary hypothesis—Reptile exceeds Independent at $K = 1$ jointly across the four targets—is confirmed: the joint stratified permutation test gives a mean paired difference of +4.3 pp (95% hierarchical-bootstrap CI +2.9 to +5.7 pp) with $p < 0.001$ over 72 partitions (Table 4, joint row); FOMAML gives +2.8 pp ($p < 0.001$); Fine-tune (+0.6 pp, $p = 0.43$) and DANN (+0.3 pp, $p = 0.73$) do not differ from Independent. At $K = 5$ the joint Reptile difference is +1.6 pp ($p = 0.01$) and at $K = 10$ and $K = 20$ it is not distinguishable from zero.

The secondary, per-basin tests are weaker, as they must be with 10–25 partitions per basin. Before correction, Reptile is significant in Copiapó ($p = 0.002$), Elqui ($p = 0.005$) and Limarí ($p = 0.007$) but not in Huasco ($p = 0.13$); after Holm correction over the eight secondary tests, Copiapó ($p_{\text{Holm}} = 0.014$), Elqui (0.028) and Limarí (0.037) remain significant and Huasco does not (0.39). Huasco is the basin in which the five seeds produce identical K-means partitions (adjusted Rand index 1.00; Table A.8), so its ten (seed, fold) partitions are two spatial folds evaluated five times; the test there has effectively two spatial replicates and cannot resolve a +2.6 pp effect. Under the most conservative replication unit, the fold with seeds averaged ($n = 5$ or fewer per basin), no per-basin test can reach $p < 0.05$ by construction (the smallest attainable Wilcoxon p is 0.0625), although the sign is positive in 5/5 folds in Copiapó, Elqui and Limarí and in 3/4 in Huasco; the joint test at this unit remains significant ($p < 0.001$).

Two further observations qualify the size of the effect. First, pooling episodes as if independent would have given confidence intervals a median 3.3× narrower than the hierarchical ones (Table 2 reports the latter). Second, an independent replication of the whole protocol with the same seeds but different network initialisations (the 2 × 2 experiment

of Section 5.4, which re-trains Independent and Reptile) gives a joint $K = 1$ difference of +2.2 pp rather than +4.3 pp, with the same ordering of basins. The variance between runs of the meta-training procedure is therefore of the same order as the effect, is not captured by the partition-level intervals (which condition on the trained meta-models), and is the reason we describe the $K = 1$ advantage as 2–4 pp rather than as a point estimate.

5.3. Feature selection with target labels inflates the transfer advantage

The submitted version of this benchmark selected features on all 14 basins and reported a $K = 1$ advantage of +5.5 to +6.2 pp. To isolate the effect of that leakage from every other change, we re-ran the 2×2 design of Section 5.4 twice with identical seeds, partitions and support draws: once with the eleven features selected using target labels and once with the nine selected on source labels only (Table A.9). The mean $K = 1$ Reptile advantage over Independent is +4.7 pp with the leaky set and +2.4 pp with the clean set; per basin, +5.0 → +2.6 (Copiapó), +3.3 → +1.5 (Huasco), +2.5 → +3.2 (Elqui) and +8.0 → +2.1 pp (Limarí). At $K \geq 5$ the two feature sets give the same advantage (1.4 vs. 1.5 pp at $K = 5$; 0.4 vs. 0.4 at $K = 10$). Absolute F_1 at $K = 1$ falls by 1–10 pp without the two leaked features, because annual precipitation and geomorphons carry genuine within-basin signal in the targets. The leaky selection also made episodic training look superior to a basin-balanced pooled baseline (+1.7, +1.5, −1.6 and +1.9 pp in the four targets), a difference that vanishes with the clean set (next section). Among the extended baselines, ProtoNet lost the most (−9.8 and −9.5 pp in Copiapó and Huasco), and pooled logistic regression the least (≤ 3 pp). Selecting features with the labels of the basins later used to test transfer is thus a fifth convention that overstates few-shot transfer: here it doubled the headline advantage.

5.4. Domain weighting versus episodic training

Episodic meta-training samples source basins uniformly, whereas the concatenated pre-training of Fine-tune weights each basin by its sample count, so that Taltal supplies 49% of the pre-training positives. The two baselines therefore differ in domain weighting as well as in optimisation. Table 5 evaluates the 2×2 design {size-weighted, basin-balanced} × {pooled, episodic} on identical partitions and support draws with five seeds. Weighting the source basins equally in pooled pre-training raises F_1 at $K = 1$ by +1.6 pp on average (+1.7, +0.7, +1.1 and +2.6 pp; significant in Elqui and Limarí). Given balanced weighting, episodic training adds nothing: Reptile minus Fine-tune (balanced) is −1.4, +0.5, +0.6 and +0.2 pp (−0.04 pp on average, no basin significant). Initialising Reptile from the pooled pre-trained weights (Pretrain→Reptile) equals Reptile from scratch (−0.1 pp) and exceeds size-weighted Fine-tune by +1.5 pp, significant in three basins. At $K = 5$ the pattern is the same at half the magnitude, and at $K \geq 10$ all contrasts are within ±0.7 pp.

The interpretation is direct. What transfers from the source pool to a new basin is a representation in which every source basin has had equal influence; episodic optimisation is one way to obtain it, basin-balanced pooled pre-training is another, and the two are indistinguishable under spatially disjoint, leakage-free evaluation. The failure of Fine-tune in the submitted version was a failure of size-weighted pooling, not of pooled pre-training

as such. This also explains the behaviour of Meta-Baseline, whose frozen encoder is size-weighted (Section 5.12).

5.5. Random versus spatial cross-validation

Random cross-validation, in which support and query pixels are drawn without regard to location (Figure 2), overstates spatial-CV F_1 by 13–21 pp in Elqui and by 5–25 pp in Limarí at every K, and by 7–9 pp at $K = 5$ in Copiapó (Figure 5). It does not do so uniformly: in Copiapó at $K = 1$ the two protocols agree within 2 pp, and in Huasco at $K = 1$ spatial CV is 6 pp *higher* than random CV, converging only at $K = 20$. The Huasco inversion follows from the fold structure documented in Section 5.6: the spatial folds that remain valid are the event-dense mountainous clusters, whereas the random query mixes the whole sampled basin, including low-relief pixels that are harder to classify from morphometry alone. The inflation by random CV is therefore real and large in the heterogeneous basins, but it is a property of each basin's spatial structure rather than a constant to be subtracted, which is one more reason to report spatial CV directly. The relative ranking of methods is preserved under both protocols, and under random CV the $K = 1$ Reptile advantage is +7.3, +4.5, +3.3 and +1.9 pp in Copiapó, Huasco, Elqui and Limarí.

5.6. Spatial independence of the folds

Spatially disjoint folds do not by themselves guarantee that query pixels are far from support pixels. We measured the separation directly (Table A.8): over the valid $K = 1$ partitions, the distance from a held-out pixel to the nearest support pixel is never below 1.9 km (Huasco), 2.2 km (Elqui), 3.9 km (Limarí) or 16.6 km (Copiapó); no held-out pixel lies within 1 km of the support, and the median separation is 17–44 km. Residual variograms of the Independent model fitted with a spherical model give autocorrelation ranges of 30 km (Copiapó), 45 km (Huasco), 21 km (Elqui) and 17 km (Limarí) (Figure B.18); the median separation exceeds the range in Copiapó and Limarí, is comparable to it in Elqui, and is below it in Huasco.

Because distance summaries are indirect, we repeated the full K-shot benchmark removing all support pixels within 2, 5, 10 and 20 km of the held-out cluster (Table A.10, Figure B.9). Buffers up to 10 km remove at most 3.5% of the support pool and leave the joint $K = 1$ Reptile advantage unchanged (+4.4, +4.2 and +4.3 pp, all $p < 0.001$). The 20 km buffer, of the order of the variogram ranges, removes 14% of the support in Huasco and Limarí and 21% in Elqui, halves the number of valid Huasco partitions, and attenuates the joint advantage to +3.6 pp ($p < 0.001$; +6.3, +0.8, +4.6 and +2.5 pp per basin). Unbuffered spatial CV therefore overstates the few-shot advantage by less than 1 pp relative to a buffer comparable to the autocorrelation range.

Two further properties of the partitions matter for interpretation. First, the K-means partitions of different seeds overlap strongly (adjusted Rand index 0.93 in Copiapó, 1.00 in Huasco, 0.84 in Elqui, 0.70 in Limarí), which is why inference is reported at the partition level and with the fold as a sensitivity unit (Section 5.2). Second, the excluded partitions are not random: every exclusion is a held-out cluster containing a single class (Copiapó 10/25, Huasco 15/25, Limarí 3/25, Elqui 0/25), and in Huasco and Limarí the excluded

clusters are the flat, event-free parts of the basin (slope standard deviation 1.7 vs. 6.7 in the retained clusters of Huasco; MRVBF 2.4 vs. 0.8; Kolmogorov–Smirnov $p < 0.001$ for eight of nine features), mapped in Figure B.8. The benchmark thus characterises transfer within the event-bearing mountainous terrain of each basin, where F_1 is defined, not across the entire catchment. Finally, re-running the benchmark with 5, 10 and 20 spatial folds (Figure B.20) shows that the $K = 1$ Reptile advantage persists at every granularity in Copiapó, Elqui and Huasco, whereas in Limarí it is +3.7 pp with 5 folds but −0.5 and +0.7 pp with 10 and 20 folds, where held-out clusters contain very few events; we therefore do not claim a fold-independent advantage in Limarí.

5.7. Prevalence-robust evaluation

Because the F_1 of the main benchmark is computed at threshold 0.5 on held-out clusters whose positive share (3–86%) is far above the prevalence of landslide pixels in a basin, it does not reflect the extreme class imbalance of operational mapping. We therefore assess the $K = 1$ advantage with two prevalence-aware lenses (Figure B.11). First, the **ROC-AUC**, which is threshold- and prevalence-independent: under spatial CV at $K = 1$, Reptile exceeds Independent in all four targets—Copiapó 0.684 vs. 0.544, Huasco 0.671 vs. 0.621, Elqui 0.693 vs. 0.606, Limarí 0.577 vs. 0.497 (+0.05 to +0.14)—so the discrimination gain is not an artifact of the operating point. Second, we translate the operating characteristics to realistic prevalence via $\text{precision}(\pi) = \pi\,\text{TPR}/[\pi\,\text{TPR} + (1-\pi)\,\text{FPR}]$ and recompute the precision–recall curve *conditional on an assumed* prevalence $\pi = 0.05$ (a plausible fraction of hazard-prone terrain; the raw pixel-level inventory prevalence, 10^{-4}–10^{-3} in these basins, is far lower, and the released result file `results/stats/prevalence_robust.csv` reports precision at $\pi \in \{0.01, 0.05, 0.10\}$). Unlike ROC-AUC, average precision depends on π and all AP values below are to be read at $\pi = 0.05$. At this prevalence *absolute precision collapses for every method*—precision at threshold 0.5 is 0.05–0.09, i.e. most flagged pixels are false positives—a direct, honest consequence of rare-event detection that the fixed-threshold F_1 masks. The meta-learning advantage survives the re-weighting: average precision at $\pi = 0.05$ favours Reptile over Independent in all four targets (Copiapó 0.097 vs. 0.050; Huasco 0.084 vs. 0.067; Elqui 0.092 vs. 0.066; Limarí 0.054 vs. 0.045). The operational reading is twofold: the framework is a terrain-*stratification* tool (its value is in ranking, captured by AUC and AP, not in per-pixel classification), and the source-learned initialisation's benefit is robust to prevalence, not an artifact of the evaluation operating point.

5.8. Susceptibility maps: what the $K = 1$ lift looks like on the ground

To translate the benchmark advantage into the product a basin manager would actually use, we apply the $K = 1$-adapted models to the *full* 30 m grid of two targets—Copiapó (largest AUC/AP gain, Section 5.7; $N = 19$) and Elqui (a hard, low-signal basin)—and map the mean adapted-model output across the three seeds and twenty support draws (Figure 6). Maps are shown as within-basin percentiles, the standard susceptibility-class presentation, because the raw $K = 1$ outputs are uncalibrated (Section 6.6). In Copiapó, both models produce spatially coherent surfaces that track the drainage network and steep valley flanks, and the 19 inventory events fall predominantly in the upper susceptibility percentiles; the

meta-learned (Reptile) surface sharpens the ridge–valley contrast and pulls the western headwater tributaries—where several inventory points cluster—into higher percentiles than the Independent baseline, consistent with its +7.4 pp F_1 edge. In Elqui, the maps make the low $F_1 \approx 0.47$ tangible: susceptibility is diffuse, the strong west–east orographic gradient dominates the pattern, and inventory events are scattered across percentile classes rather than concentrated in a coherent high-susceptibility zone. The map view thus corroborates both the headline result (the source-learned initialisation yields a usable first-pass susceptibility layer from a single seed event) and its limit (in heterogeneous basins the layer is a screening product, not a planning-grade map).

5.9. Domain adaptation: DANN and CDAN

DANN does not improve over Independent at $K = 1$ (joint +0.3 pp, $p = 0.73$; Table 4) and is the only method with a negative paired difference whose interval excludes zero in one basin (Huasco, −2.5 pp). At $K = 5$ it is weakly positive (joint +1.7 pp, $p = 0.02$), comparable to Reptile at that K, and at $K \geq 10$ it is neutral. The pattern is thus not the uniform negative transfer reported in the submitted version, which was concentrated in Copiapó under the leaky feature set, but an absence of any few-shot advantage: adversarial alignment yields nothing at the support sizes where a source-learned initialisation matters most. To verify this is not an artifact of DANN's decade-old design, we additionally evaluate **CDAN** (Long et al., 2018), a stronger conditional-adversarial method (Table 3). CDAN behaves like DANN: at $K = 1$ it reaches 0.540 (Copiapó), 0.669 (Huasco), 0.452 (Elqui) and 0.456 (Limarí), below Reptile in every target and within ±2 pp of Independent in three. That a modern conditional-adversarial method reproduces the pattern indicates the limitation is intrinsic to adversarial domain alignment under scarce source domains, not specific to the original DANN formulation. We hypothesize that adversarial alignment requires both a moderately-sized source pool *and* sufficient target signal to break ties between candidate invariances; the conjunction of few source domains ($N = 10$) and very small target supports leaves the gradient-reversal mechanism without either. We return to this in the Discussion (Section 6).

5.10. Adaptation curves and the adaptation budget

The meta-learned initialisations are discriminative before any adaptation: in Copiapó at $K = 5$ (random CV, Figure 4) FOMAML and Reptile start at $F_1 = 0.63$ and 0.62 with zero gradient steps against 0.41 for a random initialisation, reach 0.70 and 0.68 after a single step and plateau there; Fine-tune reaches 0.66 in three steps. Independent, trained from random initialization, requires > 100 steps to approach a comparable plateau. This validates the practical efficiency benefit of a source-learned initialisation: a pre-trained model is immediately deployable for new basins, whereas Independent training requires complete model retraining from random initialization.

Because the main benchmark fixes the adaptation budget at 30 inner steps for *every* method, the slower convergence of Independent raises a fairness question: does the fixed budget under-serve the baseline and thereby inflate the reported lift? A budget-fairness control answers no. Re-running the $K = 1$ spatial-CV protocol with inner-step budgets of

30, 100 and 300 for both Independent and Reptile (shared support draws per episode), Independent gains at most +1.0 pp F_1 from a 10× larger budget (Copiapó 0.512 → 0.506, Huasco 0.686 → 0.688, Elqui 0.426 → 0.436, Limarí 0.436 → 0.435): with a two-sample support, additional steps saturate or over-fit rather than converge further on held-out queries—the >100-step plateau above concerns the *trajectory* of training-loss descent, not the attainable query F_1. Measured against Independent at its own best budget, the Reptile lift remains +1.9 to +5.7 pp across the four targets (`results/stats/baseline_budget.csv`), so the fixed 30-step budget does not manufacture the meta-learning advantage. Sensitivity to the meta-training hyperparameters (inner steps $\{1, 3, 5, 10\}$, Reptile $\epsilon \in \{0.05, 0.1, 0.3\}$, $K_{\text{meta}} \in \{5, 10, 20\}$) is at most 1.4 pp F_1 (Figure B.14).

5.11. Classical ML under spatial CV

Logistic regression, random forest, XGBoost and CatBoost were evaluated under the identical K-shot spatial-CV protocol (Figure B.16). The strongest classical configuration is logistic regression trained on the pooled source basins plus the K target samples: at $K = 1$ it reaches 0.696 (Copiapó), 0.746 (Huasco), 0.472 (Elqui) and 0.503 (Limarí), equal to or above Reptile in every target (+10.3, +3.5, +0.2 and +2.9 pp). At $K \geq 5$ the picture reverses in three basins: the meta-learned MLP leads pooled logistic regression by 3–14 pp in Elqui, Huasco and Limarí, while pooled logistic regression stays ahead in Copiapó at every K; gradient-boosted trees trained on the pooled source win most Limarí cells. Pooled logistic regression was also the method least affected by the feature-selection leakage ($\leq$ 3 pp), which is consistent with a linear model extracting a transferable morphometric signal that the neural methods only match after balancing the source domains. The operational reading is therefore plain: for a new basin with one or a handful of verified events, a pooled-source logistic regression on nine DEM descriptors is at least as good as any meta-learned neural initialisation we tested, is simpler to train and audit, and needs no meta-training infrastructure. The neural-network comparison isolates a different question—how the source representation should be formed—and its answer (Section 5.4) is that balanced domain weighting, not episodic optimisation, is what matters.

5.12. Modern meta-learning baselines

To test whether the few-shot advantage is specific to first-order methods (Reptile, FO-MAML) or general to meta-learning, we evaluate two additional baselines under spatial CV with five seeds: ProtoNet (Snell et al., 2017) and Meta-Baseline (Chen et al., 2021) (Table 3, Figure 7). ProtoNet, a metric-based meta-learner with *zero* gradient adaptation at test time, matches the gradient-based meta-learners at $K = 1$ (Copiapó 0.592, Huasco 0.709, Elqui 0.441, Limarí 0.485 against Reptile's 0.594, 0.711, 0.470 and 0.474), which corroborates the interpretation of Raghu et al. (2020) that the gain comes from a transferable representation, not from the adaptation mechanism. Meta-Baseline, whose encoder is pre-trained on the size-weighted concatenation and then frozen, under-performs every other source-trained method at $K = 1$ (0.499, 0.655, 0.406, 0.422), as the domain-weighting result of Section 5.4

predicts for a size-weighted representation that receives no adaptation. Across the four targets, every episodically trained or balanced source-trained method exceeds Independent at $K = 1$; the size-weighted ones do not.

5.13. Preprocessing sensitivity

Reviewers asked how the scaling and encoding of the inputs affect models that are sensitive to numerical distance. We re-ran the $K = 1$ spatial-CV comparison of Reptile and Independent (three seeds, identical partitions and supports) under four preprocessing variants (`results/stats/preproc_sensitivity.csv`). With the per-basin, label-free standardisation used throughout the paper, the mean Reptile advantage over the four targets is +3.2 pp (+4.6, +1.7, +2.6, +3.6 in Copiapó, Huasco, Elqui, Limarí). Estimating the standardisation statistics on the source pool only, so that no target pixel contributes to preprocessing, gives +2.3 pp (+2.7, +1.3, +3.5, +1.5) with the same ordering of methods in every basin. Estimating them from the $2K = 2$ support pixels alone is degenerate (the standard deviation of two samples) and removes the advantage (−1.9 pp); it is not a usable option at $K = 1$ and we report it only for completeness. One-hot encoding the single categorical feature retained by source-only selection (landform, 11 classes; 19 inputs instead of 9) leaves Independent unchanged (0.533 vs. 0.528 mean F_1) but lowers Reptile (0.549 vs. 0.560), so that the advantage halves to +1.6 pp (+1.2, +1.3, +2.9, +0.7); integer-coded landform, which orders the classes along an approximate flat-to-steep gradient, evidently helps the meta-learned representation more than a nominal encoding does. The direction of every comparison is preserved under all three defensible variants, but the magnitude of the few-shot advantage depends on these choices by up to a factor of two, which we fold into the 2–4 pp range given in Section 5.2.

5.14. Spatial CNN backbone: spatial context substantially improves accuracy

To quantify how much the point-wise feature representation costs in predictive accuracy, we re-run the four primary methods (Independent, Fine-tune, Reptile, FOMAML) with a small spatial CNN backbone consuming 32×32 patches over the 25 gridded feature channels (Section 4.2) on the largest target basin, Huasco (Figure B.19). This experiment uses the gridded rasters directly, not the CFS-selected features, and is therefore unchanged by the feature-selection correction. **The CNN backbone outperforms the MLP by 6–11 pp F_1 across all methods and K values.** At $K = 1$, Independent CNN reaches $F_1 = 0.78$, already exceeding the MLP's best; FOMAML-CNN reaches 0.82. At $K = 20$, all CNN variants reach $F_1 \geq 0.93$ compared to ~ 0.80 for the MLP. The meta-learning lift over Independent is smaller in the CNN regime (+4.2 pp at $K = 1$ for FOMAML-CNN, with the two intervals overlapping), indicating that spatial context absorbs part of the prediction signal that the source-learned initialisation captures in the point-wise regime. Two implications follow, and one competing interpretation must be stated. First, the main-benchmark MLP results report a *conservative lower bound* on achievable cross-basin transfer accuracy; the few-shot advantage remains valid as a methodological finding but is not the operational ceiling. Second, the framework is mechanically applicable to a spatial CNN backbone, and the ordering over Independent is preserved in that regime. However, the shrinking lift also

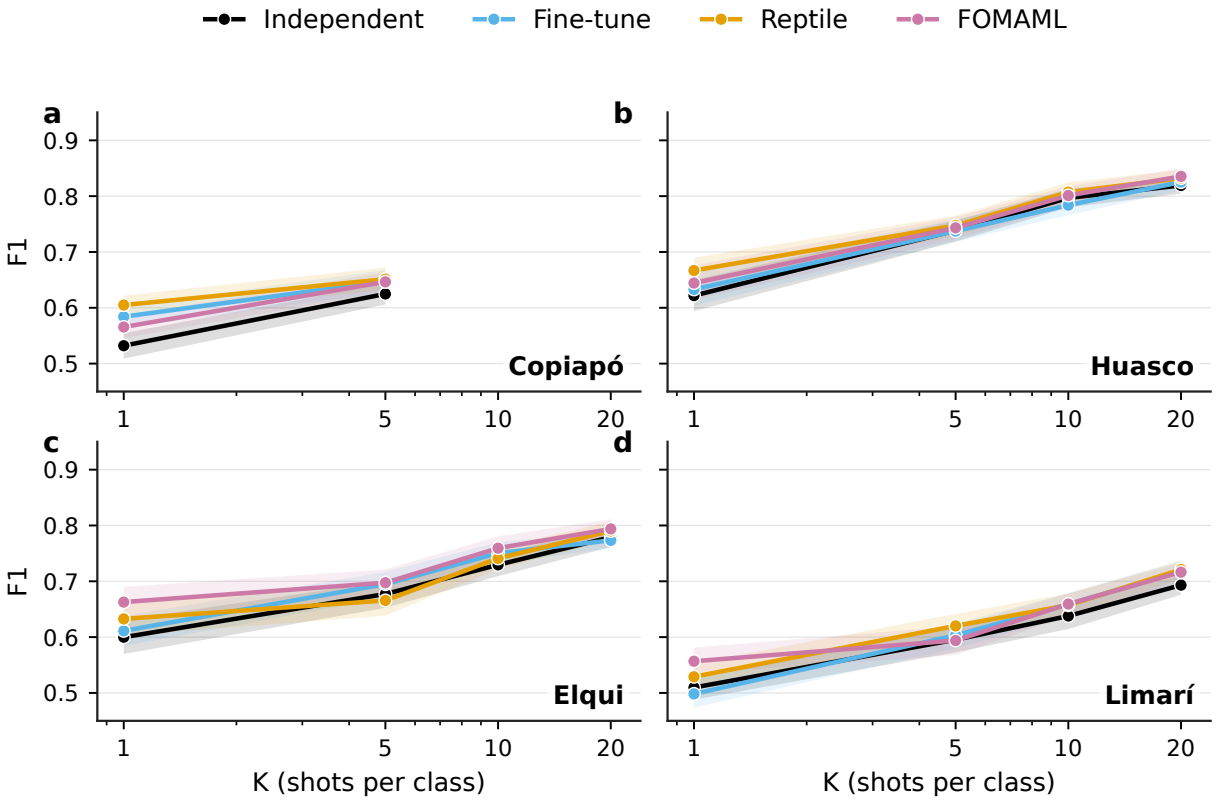


Figure 2: F_1 score versus number of support shots K under *random* cross-validation for the four target basins (source-only feature set). Lines: mean across 40 episodes × 3 seeds; shaded bands: pooled bootstrap 95% confidence intervals (random CV has no spatial cluster structure; compare the hierarchical intervals of Table 2). Random CV is the permissive foil for the spatial-CV benchmark of Figure 5 and Table 2: here Reptile leads Independent at $K = 1$ by +7.3 (Copiapó), +4.5 (Huasco), +3.3 (Elqui) and +1.9 pp (Limarí).

admits the alternative reading that *the advantage is largest precisely where the backbone is weakest*—i.e., that a stronger spatial representation absorbs the transferable signal the initialisation would otherwise contribute—and a single-basin demonstration cannot distinguish these interpretations. A multi-target CNN/U-Net meta-learning evaluation is required to resolve them and is left for follow-up work; until then, the Huasco experiment should be read as a proof of mechanism, not as evidence that the K-shot advantage carries unchanged into stronger architectures.

5.15. Precipitation similarity and transfer

Whether source–target similarity in annual precipitation ($|\text{bio_12}_{\text{source}} - \text{bio_12}_{\text{target}}|$, computed from the full feature table since bio_12 is no longer a selected feature) predicts the FOMAML advantage at $K = 5$ cannot be resolved with four targets: Pearson $r = -0.55$ ($p = 0.45$), Spearman $\rho = -0.60$ ($p = 0.40$) (Figure B.17). We report it as an underpowered null from a single-variable proxy and draw no design conclusion from it (Section 6.4).

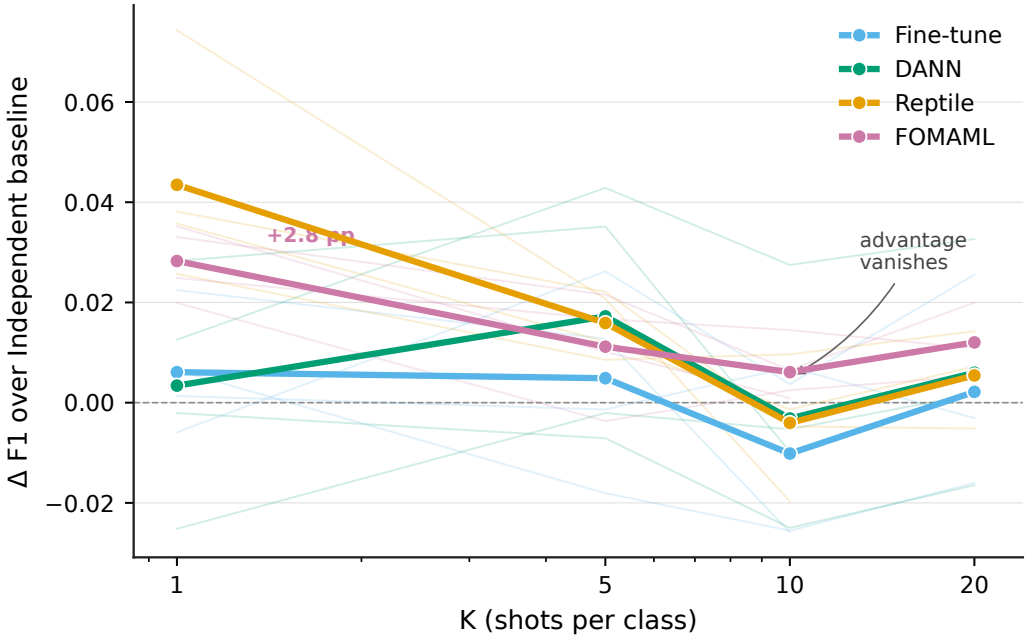


Figure 3: Mean F_1 advantage over the Independent baseline (averaged across the four target basins) under *spatial* cross-validation, source-only features, five seeds. Reptile achieves +4.3 pp at $K = 1$ (joint stratified permutation test, $p < 0.001$; Table 4), +1.6 pp at $K = 5$ ($p = 0.01$) and is indistinguishable from Independent at $K \geq 10$ (−0.4 and +0.6 pp); FOMAML follows the same decay from +2.8 pp. Fine-tune stays within ±1 pp of Independent at every K; DANN is neutral at $K = 1$ and +1.7 pp at $K = 5$. Thin lines: per-target advantages; thick lines: cross-target means.

6. Discussion

6.1. What the meta-learners learned, and what they did not

Reptile, FOMAML and ProtoNet—averaged-displacement, query-gradient and nearest-prototype meta-learners—reach statistically comparable F_1 at $K = 1$ in every target (Tables 2 and 3), and ProtoNet does so with no gradient adaptation at all. The convergence of three structurally distinct algorithms, consistent with Raghu et al. (2020), indicates that the few-shot advantage is carried by the source representation rather than by any particular adaptation rule. The 2×2 experiment of Section 5.4 then shows what property of that representation matters: equal influence of every source basin. Once pooled pre-training is given the same weighting, it is indistinguishable from episodic training; initialising episodic training from pooled weights changes nothing; and the one source-trained method that fails, Meta-Baseline, is the one whose encoder is both size-weighted and frozen. The practical consequence is that a basin-balanced pooled network, which is cheaper and simpler than any meta-learning algorithm, is the appropriate neural baseline for cross-basin susceptibility transfer, and that the customary size-weighted Fine-tune baseline should no longer be used as the reference against which meta-learning is judged. Among the meta-learners, Reptile remains our recommendation on cost grounds: its update rule is cheaper than FOMAML's query-loss pass and it needs no prototype design.

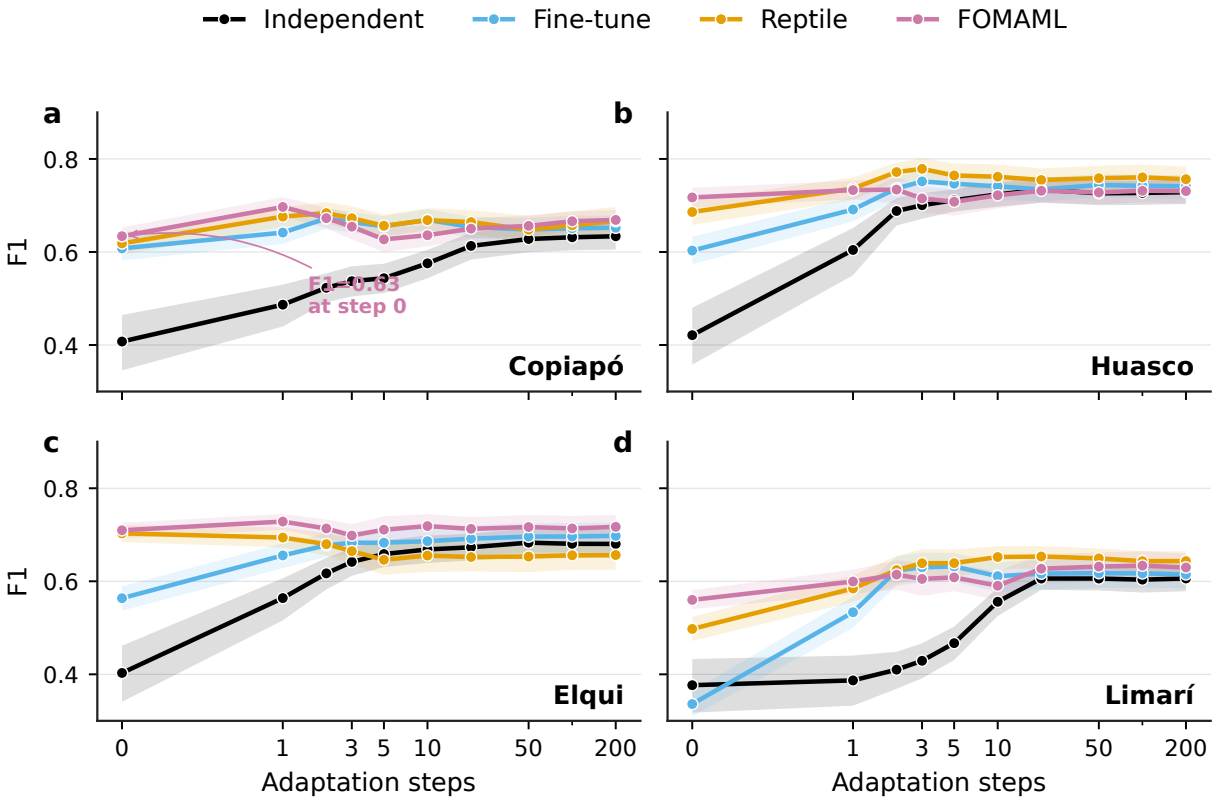


Figure 4: Adaptation efficiency at $K = 5$ across target basins (random CV, source-only features). F_1 as a function of inner Adam gradient steps applied to the support set, evaluated on held-out queries. In Copiapó the meta-learned initialisations are already discriminative before adaptation (FOMAML $F_1 = 0.63$, Reptile 0.62 at step 0, vs. 0.41 for a random initialisation) and reach ≈ 0.70 within one to three steps; Independent needs > 100 steps to approach a comparable plateau. Section 5.10 shows that this slower trajectory does not translate into a disadvantage at the fixed 30-step budget.

6.2. Why adversarial alignment does not help here

Neither DANN nor CDAN improves over Independent at $K = 1$, and DANN is weakly positive only at $K = 5$ (Section 5.9). Two mechanisms are consistent with this. First, adversarial invariance needs domain diversity: with ten source basins the domain classifier can exploit coarse descriptors such as basin-mean relief, and the gradient-reversal layer then discards signal that is also predictive of susceptibility, a form of negative transfer. Second, DANN was designed for the unsupervised setting; re-purposing its feature extractor as the starting point for a supervised K-shot fine-tune combines two mechanisms that are not jointly optimised. The robust statement is the absence of any adversarial advantage under a small source pool, not a uniform degradation: the strong Copiapó deficit reported in the submitted version did not survive the leakage-free feature set.

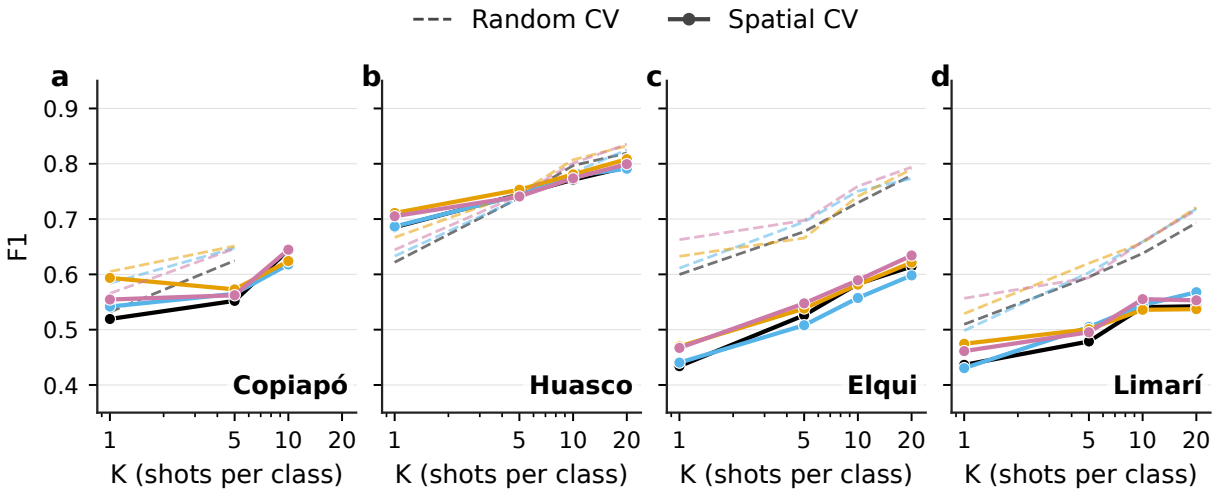


Figure 5: F_1 versus K under random (dashed) and spatial (solid) cross-validation. Spatial CV partitions each basin into 5 K-means clusters on (x, y) coordinates, so that support and query are spatially disjoint, and uses the whole held-out cluster as query. Random CV overstates F_1 by 13–21 pp in Elqui and 5–25 pp in Limarí at every K, by 0–2 pp at $K = 1$ and 7–9 pp at $K = 5$ in Copiapó, and does *not* overstate it in Huasco at $K = 1$ (-6 pp), where the retained spatial folds coincide with the event-dense mountainous terrain (Section 5.6). The relative ranking of methods is preserved.

6.3. Spatial heterogeneity in Elqui: a hard transfer case

Among target basins, Elqui exhibits the largest drop in F_1 between random and spatial CV protocols (13–21 pp at every K; Section 5.5). All methods are constrained to $F_1 \approx 0.47$–0.55 under spatial CV at any K. This indicates that intra-basin heterogeneity, not the cross-basin transfer capability, is the dominant factor in Elqui.

Two hypotheses are consistent with the data. (i) **Orographic structure**: Elqui has a strong west-east elevation gradient (sea level to $\sim$ 5,000 m), with the cordillera dominated by Mesozoic plutonic rocks (granitoids), the precordillera by Cenozoic volcanics, and the coast by Quaternary alluvial deposits. Each lithological unit produces distinct landslide styles (debris flows in the alluvium, rotational slides in the granitoids), and our nine-feature morphometric representation may not capture these axes adequately. (ii) **Trigger heterogeneity**: the SERNAGEOMIN *catastro* for Elqui aggregates events triggered by the 2015 Atacama floods, the 1997 El Niño, and the 1922 Vallenar earthquake, three regimes with distinct preparatory factors. A meta-model trained on Maule 2010 (seismic) and Chañaral (rainfall) sources cannot easily recover this trigger-specific structure from the support set alone.

Future work should address Elqui-style heterogeneity by (a) richer spatial features (e.g., multi-temporal Sentinel-1 SAR coherence), (b) trigger-aware meta-learning, or (c) fold-specific stratification. None of these are straightforward extensions, and we report Elqui as a transparent failure mode rather than attempting to mask it.

6.4. Climate and transfer: a preliminary null

Table 2: F_1 score (mean) with **two-stage hierarchical-bootstrap** 95% confidence intervals (resampling (seed, fold) partitions, then episodes; 2,000 resamples) for each target basin × shots-per-class K × method, under spatial cross-validation with the leakage-free, source-only feature set (9 features) and five seeds. Best per row in **bold**. Copiapó $K = 20$ is not reported ($N = 19$ events). Valid (seed, fold) partitions: Copiapó 15/25, Huasco 10/25, Elqui 25/25, Limarí 22/25 (Section 4.3); total realized K-shot adaptations: 5,460. Source: `results/spatial_benchmark/summary.csv`, `results/stats/hierarchical_bootstrap.csv`.

Target	K	Independent	Fine-tune	DANN	Reptile	FOMAML
Copiapó	1	0.519 [0.45,0.59]	0.542 [0.44,0.64]	0.548 [0.43,0.66]	**0.594 [0.49,0.69]**	0.555 [0.47,0.64]
	5	0.552 [0.43,0.67]	0.565 [0.42,0.68]	**0.587 [0.45,0.71]**	0.573 [0.44,0.69]	0.562 [0.43,0.67]
	10	0.644 [0.50,0.77]	0.618 [0.48,0.74]	0.634 [0.49,0.77]	0.624 [0.48,0.76]	**0.645 [0.51,0.76]**
Huasco	1	0.685 [0.65,0.72]	0.687 [0.65,0.72]	0.660 [0.62,0.69]	**0.711 [0.68,0.74]**	0.705 [0.67,0.74]
	5	0.745 [0.73,0.76]	0.743 [0.73,0.76]	0.742 [0.73,0.76]	**0.753 [0.73,0.77]**	0.741 [0.72,0.76]
	10	0.771 [0.76,0.78]	0.778 [0.76,0.79]	0.766 [0.75,0.78]	**0.781 [0.77,0.79]**	0.774 [0.76,0.78]
	20	0.794 [0.78,0.80]	0.791 [0.78,0.80]	0.796 [0.79,0.80]	**0.808 [0.80,0.82]**	0.799 [0.79,0.81]
Elqui	1	0.434 [0.36,0.51]	0.440 [0.36,0.52]	0.432 [0.35,0.51]	**0.470 [0.39,0.54]**	0.467 [0.39,0.55]
	5	0.526 [0.46,0.59]	0.508 [0.44,0.58]	0.519 [0.45,0.59]	0.538 [0.47,0.60]	**0.548 [0.48,0.61]**
	10	0.583 [0.52,0.65]	0.557 [0.48,0.63]	0.558 [0.49,0.63]	0.582 [0.51,0.65]	**0.589 [0.53,0.66]**
	20	0.614 [0.54,0.68]	0.598 [0.53,0.67]	0.598 [0.52,0.67]	0.621 [0.55,0.69]	**0.634 [0.57,0.70]**
Limarí	1	0.436 [0.37,0.50]	0.430 [0.37,0.49]	0.449 [0.40,0.50]	**0.474 [0.41,0.54]**	0.461 [0.40,0.53]
	5	0.479 [0.41,0.54]	0.505 [0.44,0.57]	**0.522 [0.46,0.58]**	0.501 [0.43,0.57]	0.495 [0.44,0.56]
	10	0.541 [0.49,0.59]	0.544 [0.49,0.60]	**0.568 [0.51,0.62]**	0.536 [0.48,0.60]	0.555 [0.50,0.61]
	20	0.542 [0.49,0.59]	0.568 [0.52,0.62]	**0.575 [0.52,0.63]**	0.537 [0.47,0.60]	0.553 [0.50,0.61]

The absence of a detectable correlation between source–target precipitation distance and transfer advantage (Section 5.15) is a null from an underpowered test (four targets, one climatic variable), not evidence of climate invariance; its sign even changed between the leaky and the clean feature sets. What the leakage-free selection does establish is narrower and firmer: no climate or geology variable is selected on the source pool, and the nine features that transfer are all DEM morphometry (landform, openness, valley- and ridge-flatness, curvedness, relief and slope variability, DEM texture). This is consistent with the interpretation that the source representation encodes generic susceptibility morphology rather than climate-specific signals, and with the loss of absolute F_1 when the within-target precipitation feature was removed. Whether climate-matched source selection would help cannot be answered with four semi-arid targets; the released benchmark is designed to make that test possible as more targets are added.

6.5. Implications for remote-sensing-driven susceptibility mapping

Three findings carry methodological weight specifically for the remote-sensing-based geohazard community. *First*, the source representation that transfers is recovered from an entirely open, globally-accessible feature stack—MERIT DEM derivatives—without proprietary inputs, optical composites or in-situ measurements. This makes the framework directly replicable in any region with global-DEM coverage. The Chilean focus is illustrative, not restrictive. *Second*, source-only feature selection (Section 3.5) retains DEM-derived terrain, relief and texture descriptors only; climate, hydrology and geology descriptors are not selected on the source pool. This has practical consequences: a 30 m DEM of moderate quality

Table 3: F_1 score (mean) with bootstrap 95% confidence intervals (pooled seed × episode; the hierarchical intervals of Table 2 are roughly 3× wider) for the three additional baselines under spatial cross-validation with the source-only feature set and five seeds. Best per row in **bold**. For comparison against the primary methods see Table 2. Source: `results/extended_baselines/summary.csv`.

Target	K	ProtoNet	Meta-Baseline	CDAN
Copiapó	1	**0.592 [0.56,0.62]**	0.499 [0.46,0.53]	0.540 [0.51,0.57]
	5	0.575 [0.55,0.61]	0.524 [0.49,0.56]	**0.595 [0.56,0.63]**
	10	0.563 [0.53,0.60]	0.551 [0.52,0.58]	**0.620 [0.59,0.66]**
Huasco	1	**0.709 [0.68,0.73]**	0.655 [0.63,0.68]	0.669 [0.64,0.69]
	5	**0.763 [0.75,0.77]**	0.735 [0.72,0.75]	0.744 [0.73,0.76]
	10	**0.775 [0.77,0.78]**	0.758 [0.75,0.76]	0.773 [0.76,0.78]
	20	0.776 [0.77,0.78]	0.763 [0.76,0.77]	**0.794 [0.79,0.80]**
Elqui	1	0.441 [0.42,0.47]	0.406 [0.38,0.43]	**0.452 [0.43,0.47]**
	5	0.521 [0.50,0.54]	0.477 [0.46,0.50]	**0.526 [0.51,0.55]**
	10	0.537 [0.52,0.56]	0.489 [0.47,0.51]	**0.560 [0.54,0.58]**
	20	0.540 [0.52,0.56]	0.505 [0.48,0.53]	**0.595 [0.58,0.61]**
Limarí	1	**0.485 [0.46,0.51]**	0.422 [0.40,0.44]	0.456 [0.43,0.48]
	5	**0.561 [0.54,0.58]**	0.477 [0.46,0.50]	0.543 [0.52,0.56]
	10	**0.565 [0.55,0.58]**	0.508 [0.49,0.52]	0.557 [0.54,0.58]
	20	0.573 [0.56,0.59]	0.541 [0.53,0.56]	**0.582 [0.57,0.60]**

(MERIT, Copernicus GLO-30, or AW3D30) is the binding constraint on deployment, not optical-sensor availability. *Third*, the advantage of a source-learned initialisation is largest at $K = 1$, which corresponds to one verified mass-movement footprint visible in high-resolution Sentinel-2 or Planet imagery. A remote-sensing analyst can typically validate a single landslide via visual interpretation in < 10 minutes, whereas mapping the $K \geq 50$ events required by per-basin training takes weeks. The human-in-the-loop cost thus moves from inventory construction to seed-event validation, with two caveats that the present results impose. The first is scope: given a pre-trained source model and a pre-computed feature stack, adapting to a new basin takes minutes of compute, but assembling the DEM-derived stack for a new basin is a separate preprocessing step of hours to a day, and the few-shot advantage over a pooled logistic regression on the same stack is nil. The second is quality: at operational prevalence the per-pixel precision of every method is 0.05–0.09, so the layer produced from one event is a terrain-stratification product for screening and prioritisation, not a planning-grade map, and in heterogeneous basins such as Elqui it inherits the low discrimination of Table 2. In the Chilean institutional context, such screening layers are consumable by the agencies that motivated this study: SERNAGEOMIN, which maintains the national *catastro de remociones en masa* used here as the label source, and SENAPRED, whose regional risk-management plans require basin-scale susceptibility inputs. A natural extension is to incorporate multi-temporal Sentinel-1 InSAR coherence and amplitude features (Poggi et al., 2026; Biggs et al., 2026; Elliott et al., 2026), InSAR-augmented inventories of the kind built by Chen et al. (2023), and the multi-decadal dynamic inventories of Ma et al. (2025), to capture surface-deformation precursors that static DEM stacks cannot, an avenue actively

Table 4: Cluster-level paired comparison of each method against the Independent baseline at $K = 1$ under spatial cross-validation (source-only features, five seeds). The paired unit is the (seed, fold) spatial partition (n_c per target); the paired difference is the per-cluster mean-F_1 gap. Intervals are 95% hierarchical-bootstrap CIs (2,000 resamples); *CI excludes zero. p is the two-sided Wilcoxon signed-rank test on the cluster differences; p_{Holm} applies the Holm correction over the eight pre-specified secondary tests (four targets × {Reptile, FOMAML}). The **primary, pre-specified hypothesis** is the joint row: a permutation test stratified by target (sign flips of the cluster differences within each target, 20,000 permutations; statistic = mean of per-target mean differences). Source: `results/stats/cluster_paired_vs_independent.csv`, `secondary_holm_r1.csv`, `joint_tests_r1.csv`.

Target	**Method**	**ΔF_1 vs Indep. (pp) [95% CI]**	**Clusters > 0**	**Wilcoxon p**	**p_{Holm}**
Copiapó ($n_c = 15$)	Reptile	+7.4 [+3.9, +11.0]*	13/15	0.002	0.014
	FOMAML	+3.5 [-0.1, +7.8]	10/15	0.135	0.393
	Fine-tune	+2.2 [-2.4, +6.7]	10/15	0.454	—
	DANN	+2.8 [-2.4, +8.4]	8/15	0.421	—
Huasco ($n_c = 10$)	Reptile	+2.6 [+0.2, +4.9]*	7/10	0.131	0.393
	FOMAML	+2.0 [-0.5, +4.7]	7/10	0.232	0.393
	Fine-tune	+0.1 [-1.7, +2.0]	5/10	1.000	—
	DANN	−2.5 [-5.0, -0.3]*	2/10	0.105	—
Elqui ($n_c = 25$)	Reptile	+3.6 [+1.3, +5.9]*	17/25	0.005	0.028
	FOMAML	+3.3 [+1.2, +5.7]*	17/25	0.015	0.059
	Fine-tune	+0.6 [-1.3, +2.8]	11/25	0.812	—
	DANN	−0.2 [-2.9, +2.8]	11/25	0.634	—
Limarí ($n_c = 22$)	Reptile	+3.8 [+1.6, +6.1]*	15/22	0.007	0.037
	FOMAML	+2.5 [+1.2, +3.8]*	16/22	0.002	0.013
	Fine-tune	−0.6 [-2.8, +1.7]	10/22	0.503	—
	DANN	+1.3 [-1.8, +4.3]	14/22	0.406	—
Joint test across the four targets (stratified permutation; primary hypothesis in bold)					
Joint ($n_c = 72$)	**Reptile**	+4.3 [+2.9, +5.7]	—	<0.001	—
Joint ($n_c = 72$)	FOMAML	+2.8 [+1.5, +4.2]	—	<0.001	—
Joint ($n_c = 72$)	Fine-tune	+0.6 [-0.9, +2.1]	—	0.431	—
Joint ($n_c = 72$)	DANN	+0.3 [-1.5, +2.1]	—	0.734	—

explored for global InSAR landslide monitoring (Guo et al., 2026).

6.6. Limitations

We acknowledge ten limitations.

Geographic coverage of the targets. Transfer is evaluated on four semi-arid basins of north-central Chile (27–31°S). The hyper-arid-to-humid gradient is present in the source pool, not in the target set; transfer to humid, Patagonian or extra-Andean basins is untested, and the conclusions are stated for the evaluated targets.

Feature selection in the submitted version. The originally submitted benchmark selected features on all 14 basins, so target labels influenced the feature space. We corrected this to source-only selection and quantified the effect (Section 5.3); the correction removed two

Table 5: Domain weighting versus episodic training at $K = 1$ under spatial cross-validation (source-only features, five seeds, identical partitions and support draws for all five initialisations). Top: mean F_1 per target. Bottom: cluster-level paired contrasts (mean difference in pp F_1 with 95% hierarchical-bootstrap CI; *CI excludes zero; two-sided Wilcoxon p on the n_c partition differences) isolating the effect of basin-balanced weighting in pooled pre-training, of episodic training given balanced weighting, and of adding an episodic phase after pooled pre-training. Source: `results/pooled_vs_episodic/{summary,cluster_paired}.csv`.

		Copiapó	**Huasco**	**Elqui**	**Limarí**
Mean F_1 at $K = 1$					
Independent		0.522	0.680	0.450	0.433
Fine-tune		0.552	0.679	0.457	0.424
Fine-tune (balanced)		0.570	0.687	0.468	0.451
Reptile		0.556	0.691	0.474	0.453
Pretrain→Reptile		0.566	0.688	0.471	0.446
Contrast (pp)		**Copiapó**	**Huasco**	**Elqui**	**Limarí**
Balanced − concat pooled (domain weighting)	Δ	+1.7 [-1.6, +5.1] $p = 0.45$, $n_c = 15$	+0.7 [-0.2, +1.6] $p = 0.23$, $n_c = 10$	+1.1 [+0.2, +2.1]* $p < 0.01$, $n_c = 25$	+2.6 [+0.8, +4.5]* $p = 0.02$, $n_c = 22$
Reptile − balanced pooled (episodic vs. pooled)	Δ	−1.4 [-5.2, +1.9] $p = 0.92$, $n_c = 15$	+0.5 [-0.0, +0.9] $p = 0.16$, $n_c = 10$	+0.6 [-0.6, +1.6] $p = 0.13$, $n_c = 25$	+0.2 [-1.8, +2.0] $p = 0.92$, $n_c = 22$
Pretrain→Reptile − concat pooled	Δ	+1.4 [-0.2, +2.8] $p = 0.06$, $n_c = 15$	+0.9 [+0.4, +1.5]* $p < 0.01$, $n_c = 10$	+1.4 [+0.4, +2.4]* $p < 0.01$, $n_c = 25$	+2.1 [+0.8, +3.6]* $p < 0.01$, $n_c = 22$
Pretrain→Reptile − Reptile	Δ	+1.0 [-1.0, +3.2] $p = 0.60$, $n_c = 15$	−0.3 [-1.1, +0.7] $p = 0.49$, $n_c = 10$	−0.3 [-0.9, +0.2] $p = 0.58$, $n_c = 25$	−0.7 [-3.0, +1.7] $p = 0.59$, $n_c = 22$
Reptile − Independent	Δ	+3.3 [-0.3, +6.7] $p = 0.08$, $n_c = 15$	+1.1 [+0.1, +2.1]* $p = 0.05$, $n_c = 10$	+2.4 [+0.9, +3.9]* $p < 0.01$, $n_c = 25$	+2.0 [+0.2, +4.0]* $p = 0.09$, $n_c = 22$

features with genuine within-target signal, lowered absolute F_1 by 1–10 pp at $K = 1$ and halved the headline advantage. We report both runs in Table A.9.

Replication structure. Seeds do not generate independent spatial partitions: K-means on coordinates returns identical partitions in Huasco and strongly overlapping ones elsewhere (ARI 0.70–0.93). Inference is therefore reported at the partition level and, as a sensitivity, at the fold level, where no per-basin test can be significant with five folds; the joint test is the primary criterion. In addition, the between-run variance of meta-training (+4.3 vs. +2.2 pp joint advantage in two runs of the same protocol) is not captured by partition-level intervals, which condition on the trained meta-models; more seeds, or a bootstrap over meta-models, would be needed to include it. The magnitude also depends on preprocessing: one-hot encoding the categorical landform feature halves the three-seed estimate of the $K = 1$ advantage (Section 5.13).

Selection by excluded partitions. Held-out clusters without both classes are skipped; the excluded clusters are the low-relief, event-free parts of Huasco and Limarí (Table A.8). The benchmark therefore describes performance in the event-bearing mountainous terrain of each basin, where F_1 is defined, and is silent about the flat terrain where the models would mostly predict absence.

Source domain count. With 10 source basins, the regime is at the lower end of typical few-shot benchmarks (Omniglot uses 1,623 classes, miniImageNet uses 64 source classes), and one polygon inventory (Taltal) supplies 49% of the source positives. Basin-balanced weighting (Section 5.4) neutralises the dominance for pooled pre-training; whether a larger

Table 6: Sensitivity of the $K = 1$ meta-learning advantage to non-landslide (negative) sample selection, under spatial cross-validation (mean F_1 over three seeds; source-only features). Target negatives are re-generated by random sampling outside exclusion buffers of 500 m, 1 km and 2 km, and by a slope-matched strategy whose negatives reproduce the positives' slope distribution (removing the steep-vs-flat confound). The meta-learned initialisation (Reptile) reuses the unchanged source meta-model. Features are re-extracted from rasters, so absolute F_1 differs slightly from Table 2. Absolute F_1 shifts with the sampling choice (consistent with Abdelkader et al. 2025), but the meta-learning *lift* stays positive throughout (range +2.0 to +6.8 pp) and *increases* under the slope-matched condition. Source: `results/stats/negatives_sensitivity.csv`.

Target	**Negative-sampling strategy**	**Indep.** F_1	**Reptile** F_1	**Lift (pp)**
Copiapó ($N = 19$)	Random, 500 m buffer (baseline)	0.513	0.533	+2.0
	Random, 1 km buffer	0.526	0.588	+6.2
	Random, 2 km buffer	0.536	0.605	+6.8
Huasco ($N = 278$)	Random, 500 m buffer (baseline)	0.620	0.643	+2.3
	Random, 1 km buffer	0.679	0.698	+2.0
	Random, 2 km buffer	0.668	0.691	+2.3
	Slope-matched	0.582	0.637	+5.5

and more even source pool would increase the advantage of any method is untested. Adding basins from neighboring countries (Peru, Argentina, Bolivia, Ecuador) is a natural next step.

MLP backbone. The main benchmark uses a 3-layer MLP on point-wise tabular features. This is intentionally conservative: it ensures interpretability, avoids large-scale GPU requirements, and maps cleanly to the practical workflow of basin managers. The U-Net-style CNN baseline of Section 5.14 quantifies the cost of this choice on Huasco: the spatial CNN improves F_1 by 6–11 pp across methods and K values, and the meta-learning lift is partially absorbed by the spatial representation. The MLP results should therefore be read as a conservative lower bound on achievable accuracy. A multi-target U-Net meta-learning evaluation, including richer patch sizes and attention-based architectures, is the natural follow-up.

Static climate. Our climate features (WorldClim 2.1 bioclimatics) represent a 1970–2000 baseline, and none of them is retained by source-only feature selection; climate therefore enters the models only through the source pool composition. Climate-change-driven shifts in precipitation regimes are a known driver of changing landslide hazard but are absent from our model. Coupling with CMIP6 projections is straightforward in principle.

Probability calibration and prevalence. Two coupled caveats govern how the outputs should be read. First, the models are evaluated at a fixed 0.5 threshold on held-out clusters whose positive share is far above operational prevalence; under the extreme class imbalance of real basins, absolute precision is 0.05–0.09 (Section 5.7, Figure B.11), and the average-precision figures are conditional on the assumed $\pi = 0.05$. Second, a model adapted on only $2K$ support samples has no calibration guarantee, so its raw sigmoid outputs are not probabilities in the frequentist sense—which is why the susceptibility maps (Figure 6) are shown as within-basin percentiles. Both caveats point to the same operational conclusion:

the framework should be used as a *ranking/stratification* tool, evaluated with prevalence-independent metrics (ROC-AUC) and prevalence-conditional ones stated as such, rather than as a per-pixel binary classifier. Where a calibrated probability is required, post-hoc calibration on the K-shot support (Platt scaling or isotonic regression) is the natural next step, at the cost of consuming part of the already-scarce support budget.

Negatives generation. Twelve of fourteen basins lack explicit negative samples, requiring us to generate them by random sampling within the basin polygon outside a 500 m buffer of positives (the rule of each basin is listed in Table A.7). This introduces an implicit assumption—that all unlabeled pixels are negatives—which is statistically defensible (landslide footprints are sparse) but methodologically simplistic, and non-landslide sample selection is known to affect absolute susceptibility performance (Abdelkader et al., 2025). Heterogeneous negative rules across source basins could also act as a domain signal. To test whether our conclusion depends on this choice, we re-ran the $K = 1$ spatial-CV benchmark on the two exemplar targets (Huasco, Copiapó) under three exclusion-buffer widths (500 m, 1 km, 2 km) and a slope-matched strategy that reproduces the positives' slope distribution (Table 6). As expected, absolute F_1 shifts with the sampling scheme, but the Reptile *lift* over Independent remains positive in every condition (range +2.0 to +6.8 pp) and, tellingly, *grows* under the slope-matched scheme (+5.5 pp in Huasco): when the easy "steep-versus-flat" confound is removed, the source-learned initialisation helps more, not less, indicating the advantage is not an artifact of easy random negatives. Positive-Unlabeled (PU) learning remains a more rigorous alternative to the all-unlabeled-are-negative assumption.

Mixed triggers and movement types. The inventories combine rainfall-triggered flows (Chañaral, Taltal, the northern catastro basins), earthquake-triggered slides of four typologies (Maule 2010) and mixed-trigger records (Elqui, with events from 1922, 1997 and 2015), treated as one binary susceptibility class. Susceptibility, as distinct from hazard, concerns where slopes fail given static conditioning factors, and national inventories are routinely aggregated for that purpose; but a seismically triggered source inventory may encode conditioning patterns that do not transfer to rainfall-dominated targets, and the trigger of most catastro records is not recorded. Trigger-aware sampling and feature design would likely improve discrimination, particularly in Elqui, and a leave-one-source-out analysis by trigger is the appropriate next test.

6.7. Practical recommendations for new-basin susceptibility modeling

For a researcher or basin manager facing a *newly-studied* watershed with fewer than 10 documented landslide events, our findings support the following protocol:

1. **Start from a pooled-source model with every source basin weighted equally.** A pooled logistic regression on open DEM descriptors matched or exceeded every neural method at $K = 1$ in all four targets; a basin-balanced pooled network is the appropriate neural alternative. Size-weighted pooling, in which one large inventory dominates, is the configuration to avoid.

2. **Reach for a meta-learned initialisation when adaptation must happen in a few gradient steps** and a neural backbone is required; Reptile is the cheapest of the meta-learners we tested and statistically equivalent to FOMAML and ProtoNet. Expect a 2–4 pp F_1 advantage over training from scratch at one event per class, none beyond ten.
3. **Avoid adversarial domain adaptation** when source-domain count is small ($N \leq 10$); it offered no few-shot advantage here.
4. **Select features and estimate every data-dependent parameter on the source basins only.** Including the target labels in feature selection doubled the apparent few-shot advantage in this benchmark.
5. **Validate with spatial cross-validation, not random hold-out**, report the separation between query and support pixels, and check an exclusion buffer of the order of the residual autocorrelation range: here buffers up to 10 km changed nothing and 20 km cost 0.7 pp. Random CV inflated F_1 by up to 21 pp in the heterogeneous basins and by nothing in another; the *relative* method ranking was preserved, but the *absolute* numbers reported to stakeholders should come from spatial CV.
6. **Report uncertainty at the level of spatial partitions, with a pre-specified joint hypothesis**; pooled episode-level intervals were 3.3× too narrow, per-basin tests with ten partitions are underpowered, and seeds do not create independent partitions.
7. **Adapt for ≤ 30 gradient steps**. Source-learned initializations reach near-peak F_1 within 1–3 steps; further adaptation provides diminishing returns and risks over-fitting on the small support set.
8. **Document hard cases**. Basins with strong intra-basin heterogeneity (multiple lithologies, multiple triggers, strong elevation gradient) may resist transfer and require trigger-stratified or lithology-stratified analysis as a follow-up.

7. Conclusions

We asked how much the standard conventions for evaluating cross-basin landslide susceptibility transfer overstate it, and what survives when they are removed, using few-shot meta-learning—whose capacity to adapt susceptibility models from few samples was established by Chen et al. (2022) and Ma et al. (2025)—across 14 Chilean watersheds—ten source basins spanning the Atacama-to-Patagonia gradient and four semi-arid target basins of north-central Chile—as the test case. Five conventions were isolated on identical data. Random cross-validation inflates F_1 by up to 21 pp in the heterogeneous basins and not at all in one basin. Balanced evaluation at a fixed threshold hides that precision at a 5% prevalence is 0.05–0.09 for every method. Pooling correlated episodes narrows confidence intervals 3.3×. Selecting features with the labels of the target basins doubled the few-shot advantage of meta-learning, from +2.4 to +4.7 pp at $K = 1$ under identical seeds, and was the reason the submitted version of this work reported a larger effect. And comparing episodic meta-training against a size-weighted pooled baseline, as is customary, attributes to the meta-learning algorithm an advantage that belongs to domain weighting.

Under the strict protocol—source-only feature selection, spatially disjoint and buffered validation, partition-level inference with a pre-specified joint hypothesis, five seeds—three findings remain:

1. **A source-learned initialisation helps in the extreme few-shot regime, and the effect is modest.** Reptile exceeds training from scratch at $K = 1$ by +4.3 pp F_1 jointly across the four targets ($p < 0.001$; +2.6 to +7.4 pp per basin, significant in three of four after Holm correction, not in Huasco where the spatial partitions coincide), by +1.6 pp at $K = 5$, and by nothing at $K \geq 10$. The gain in ROC-AUC (+0.05 to +0.14) and in prevalence-weighted average precision survives re-weighting to realistic prevalence, and the advantage is insensitive to exclusion buffers of up to 10 km and attenuates to +3.6 pp at 20 km. An independent replication of the protocol gives +2.2 pp, and one-hot encoding the one categorical input halves the three-seed estimate, so the honest statement is an advantage of about 2–4 pp at one labelled event per class, whose magnitude depends on run-to-run and preprocessing choices by a factor of two.
2. **Domain weighting, not episodic optimisation, is what transfers.** Pooled pre-training with every source basin weighted equally reproduces the meta-learning advantage; given balanced weighting, episodic training and pooled training are indistinguishable (−0.04 pp over four targets and five seeds), and a frozen size-weighted encoder fails. The customary Fine-tune baseline underperformed because one inventory supplied half of its training data, not because pooled pre-training cannot transfer. A pooled-source logistic regression on nine DEM descriptors matches or exceeds every neural method at $K = 1$ in all four basins.
3. **Domain-adversarial alignment offers no few-shot advantage** with ten source domains: DANN and CDAN equal Independent at $K = 1$ and DANN is weakly positive only at $K = 5$.

Limitations include the use of a small tabular MLP backbone (a spatial CNN improves F_1 by 6–11 pp on one basin), the restriction of the evaluated transfer to four semi-arid targets, mixed triggers and movement types in the inventories, one polygon inventory supplying half of the source positives, and a between-run variance of the meta-training procedure that is of the same order as the effect. Future work should extend the evaluation to humid and Patagonian targets and to U-Net architectures across several basins, and should treat trigger-aware meta-learning as the natural response to intra-basin heterogeneity of the Elqui kind.

For practitioners, the recommendation is deliberately modest and now rests on a comparison that controls for the confounds above: start a new basin from a pooled-source logistic regression or a basin-balanced pooled network on open DEM descriptors, validate with spatially disjoint and buffered folds, report uncertainty at the partition level, and expect a first-pass susceptibility layer that ranks terrain usefully but whose per-pixel precision at operational prevalence is low. The methodological lesson generalises beyond landslides: a cross-site transfer benchmark should be built so that the test sites contribute nothing—no

labels, no features, no weighting—to the model before the moment of evaluation, and should state how much each relaxation of that rule would have added to the reported skill. The released benchmark, datasets and protocol are intended to make that standard reproducible in other data-scarce regions of Latin America and beyond.

CRediT author contribution statement

Francisco Parra: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Verónica Gil-Costa**: Methodology, Validation, Writing – review & editing, Supervision. **Carolina Bonacic**: Methodology, Validation, Writing – review & editing. **Mauricio Marín**: Conceptualization, Resources, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research was supported by the Postdoctoral Project DICYT 062619MC_POSTDOC, Universidad de Santiago de Chile. The funding source had no involvement in the study design; collection, analysis or interpretation of data; writing of the report; or the decision to submit the article for publication.

Acknowledgments

The authors thank SERNAGEOMIN (Servicio Nacional de Geología y Minería de Chile) for public access to the national *catastro de remociones en masa*, and Serey et al. (2019) for making the 2010 Maule earthquake landslide inventory publicly available. Map lines in this article delineate study areas and do not necessarily depict accepted national boundaries.

Data and code availability

- Code (pipeline scripts, figure-generation scripts, LaTeX source, aggregated benchmark summaries): `https://github.com/franciscoparrao/paper4-meta-landslide`.
- Per-basin ML-ready H5 datasets and CNN patch tensors are regenerated from public sources (SERNAGEOMIN *catastro de remociones en masa*; MERIT-DEM; WorldClim 2.1; SERNAGEOMIN geology; DGA Cuencas-BNA polygons) following the instructions in the repository README; a Zenodo deposit of the harmonised intermediate products is planned upon acceptance.
- SurtGIS toolkit, used for terrain and hydrology factor computation, is described in Parra (2026) and available at `https://github.com/franciscoparrao/surtgis`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Code (Anthropic) to assist with prose drafting, code refactoring, statistical-result tabulation, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Supplementary tables

Tables A.7–A.10 document, respectively, the inventory and negative-sampling rule of every basin, the diagnostics of the spatial-CV partitions, the two feature-selection runs, and the buffered spatial cross-validation.

Appendix B. Supplementary figures

Table A.7: Basin-by-basin inventory and sampling summary (reviewer request). Positive pixels are inventory points snapped to the 30 m grid (Taltal: polygons rasterised, ≤ 30 pixels per polygon). Explicit negatives exist in two basins only (Chañaral: 910 mapped non-landslide points; Taltal: 300 points expanded by a 60 m buffer and completed with random samples); the other twelve basins use random sampling outside a 500 m exclusion buffer around positives. All basins are balanced 1:1. 'Dominant trigger' is taken from the inventory source and is indicative only. Source: `results/stats/basin_summary.csv`.

Basin	Role	Inventory	Events	Expl. neg.	Pos. px	Neg. px	Dominant trigger	Share src. pos.
Taltal	source	Parra (2025) thesis; Taltal/Q. Negra–Pan de Azúcar	80	300	2003	2003	rainfall (March 2015)	0.485
Chanaral	source	Parra (2025) thesis suppl.; Chañaral/Río Salado	800	910	562	562	rainfall (March 2015 Atacama floods)	0.136
Rapel	source	SERNAGEOMIN catastro	392	0	387	387	mixed (catastro)	0.094
Magallanes	source	SERNAGEOMIN catastro	413	0	361	361	mixed (catastro, rainfall/snow)	0.087
Maipo	source	SERNAGEOMIN catastro	349	0	335	335	mixed (catastro, rainfall/snowmelt)	0.081
Maule	source	Serey et al. (2019) Mw 8.8 2010 inventory, 4 typologies	203	0	203	203	earthquake (2010 Maule)	0.049
Tilviche	source	SERNAGEOMIN catastro	127	0	127	127	mixed (catastro, rainfall-dominated)	0.031
Choapa	source	SERNAGEOMIN catastro	62	0	58	58	mixed (catastro)	0.014
Caracoles	source	SERNAGEOMIN catastro	50	0	48	48	mixed (catastro)	0.012
Bueno Puelo	source	SERNAGEOMIN catastro	47	0	43	43	mixed (catastro, rainfall)	0.010
Copiapó	target	SERNAGEOMIN catastro	21	0	19	19	mixed (catastro; 2015 floods)	—
Elqui	target	SERNAGEOMIN catastro	141	0	132	132	mixed (2015 floods, 1997 El Niño, 1922 Vallenar earthquake)	—
Huasco	target	SERNAGEOMIN catastro	296	0	278	278	mixed (catastro; 2015 floods)	—
Limarí	target	SERNAGEOMIN catastro	42	0	35	35	mixed (catastro)	—

Table A.8: Spatial-CV partition diagnostics per target basin (K-means, 5 folds, seeds 42/123/7/1729/2718, as in the main benchmark). 'Valid' partitions at $K = 1$ are those whose held-out cluster contains both classes; all exclusions are single-class held-out clusters. ARI is the mean [min, max] adjusted Rand index between the partitions of different seeds (1 = identical partitions). NN distance is the distance from each held-out pixel to the nearest pixel of the training clusters, summarised over valid partitions. Held-out size and positive share are the ranges over valid partitions. Source: `results/stats/fold_validity.csv`, `fold_overlap.csv`, `fold_nn_distance.csv`.

Target	Valid / total	Excluded (single-class)	Held-out size (px)	Positive share	ARI across seeds	NN min (km)	NN p5 (km)	NN median (km)
Copiapó	15/25	10	7–12	0.57–0.73	0.93 [0.88, 1.00]	16.6	26.4	43.6
Huasco	10/25	15	119–265	0.50–0.83	1.00 [1.00, 1.00]	1.9	6.4	16.7
Elqui	25/25	0	23–107	0.03–0.74	0.84 [0.72, 0.94]	2.2	7.4	17.2
Limarí	22/25	3	6–25	0.14–0.86	0.70 [0.45, 0.92]	3.9	12.6	27.5

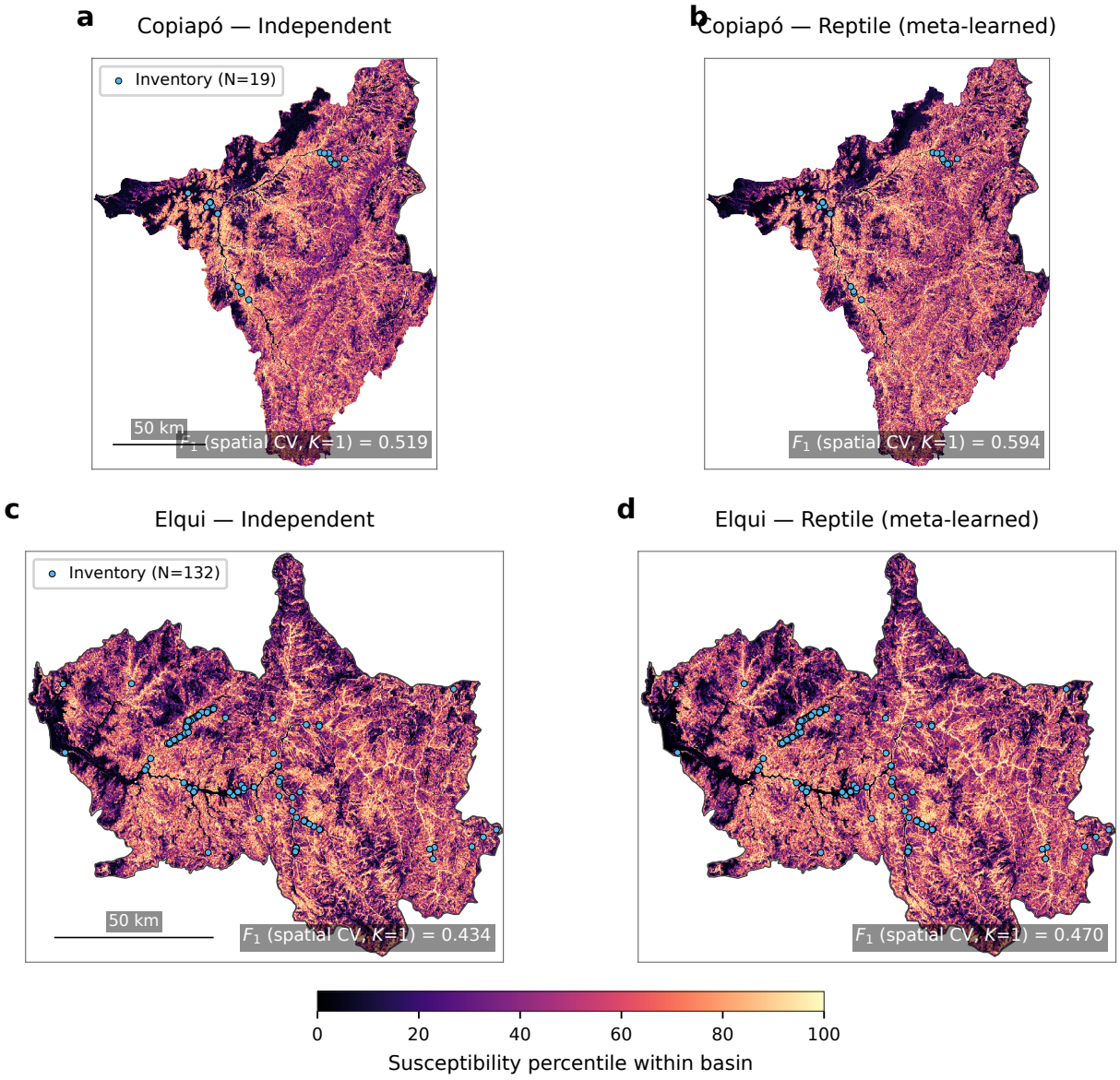


Figure 6: Full-basin susceptibility maps from $K = 1$-adapted models under the spatial-CV protocol, for Copiapó (top; smallest inventory, $N = 19$, largest AUC and average-precision gain, Section 5.7) and Elqui (bottom; a hard low-signal basin). Columns: Independent baseline (**a**, **c**) versus the meta-learned Reptile initialization (**b**, **d**), each the mean output over 3 seeds × 20 support draws (source-only feature set), shown as within-basin susceptibility percentiles (raw $K = 1$ outputs are uncalibrated; Section 6.6). Blue points: SERNAGEOMIN inventory events. The Reptile surface sharpens the ridge–valley contrast and lifts the inventory-bearing western tributaries of Copiapó into higher percentiles, consistent with its +7.4 pp F_1 edge; in Elqui the diffuse pattern and scattered inventory make the low $F_1 \approx 0.47$ (Table 2) visible on the ground. The 50 km scale bar and per-panel spatial-CV F_1 ($K = 1$) are annotated.

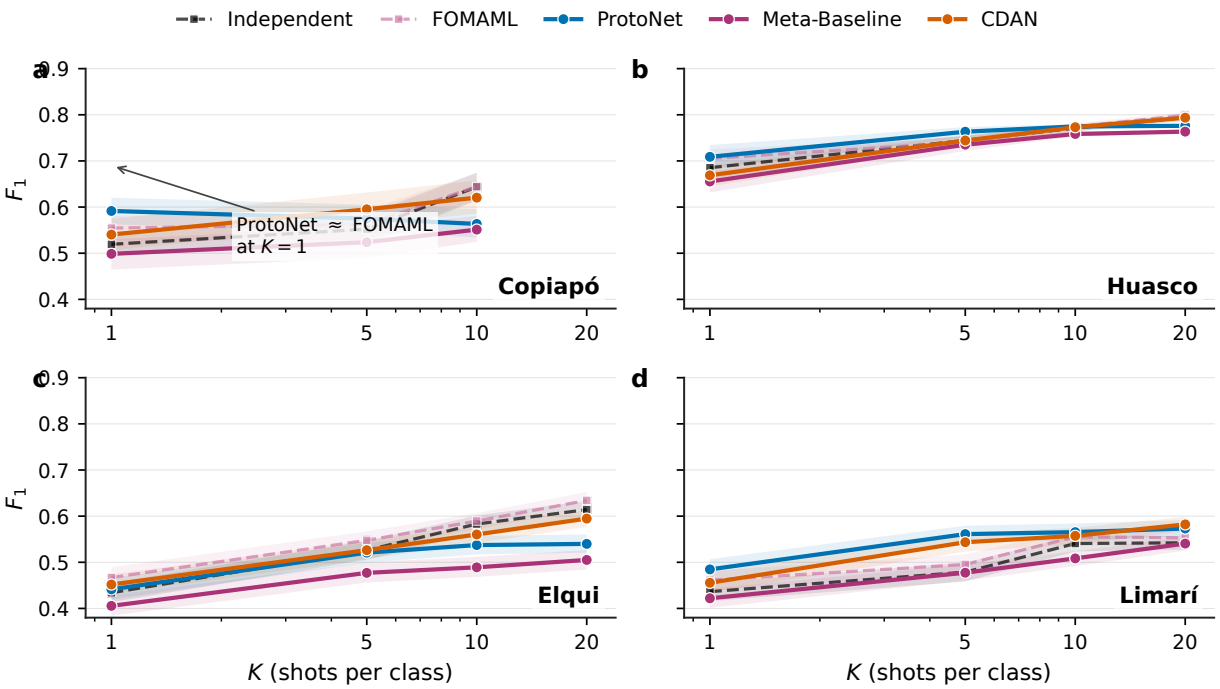


Figure 7: Extended baselines under spatial cross-validation (five seeds, source-only features): ProtoNet, Meta-Baseline, and CDAN (solid lines) compared against the reference Independent baseline and FOMAML (dashed). ProtoNet, a metric-based meta-learner requiring no gradient adaptation, matches the gradient-based meta-learners at $K = 1$ (Copiapó 0.59, Huasco 0.71). Meta-Baseline (frozen, size-weighted concat-pre-trained encoder) consistently lags. Shaded bands are pooled bootstrap 95% confidence intervals.

Table A.9: Correlation-based feature selection run on all 14 basins (the submitted version, which used target-basin labels) versus on the 10 source basins only (this revision). Step = order of entry in the forward search; SU = symmetric uncertainty with the class label. The source-only run selects 9 of the original 11 features; geomorphons and annual precipitation (bio_12) are not selected. Source: `results/stats/cfs_comparison.csv`.

Feature	Step (all 14)	SU (all 14)	Step (source 10)	SU (source 10)
`terrain__landform`	1	0.1827	1	0.1828
`texture__glcm_entropy_dem`	4	0.1475	2	0.1688
`terrain__mrvbf`	3	0.1432	3	0.1609
`terrain__openness_positive`	5	0.1657	4	0.1775
`focal_stats__dem_std_r10`	6	0.1628	5	0.1767
`focal_stats__slope_std_r4`	2	0.1616	6	0.1614
`texture__glcm_correlation_dem`	8	0.1397	7	0.1575
`terrain__mrrtf`	10	0.0449	8	0.0589
`terrain__curvedness`	7	0.1429	9	0.1481
`terrain__geomorphons`	9	0.1391	—	—
`climate__bio_12`	11	0.0799	—	—

Table A.10: Buffered spatial cross-validation at $K = 1$ (source-only features, five seeds). Support pixels within the stated distance of the held-out cluster are removed from the support pool before sampling. Cells give the cluster-level paired Reptile − Independent difference in pp F_1 with 95% hierarchical-bootstrap CI and the number of valid partitions n_c; the joint row is the stratified permutation test across targets. Residual-variogram ranges: Copiapó 30, Huasco 45, Elqui 21, Limarí 17 km. Source: `results/stats/buffered_cv_summary.csv`.

Target	0 km	2 km	5 km	10 km	20 km
Copiapó	+7.4 [+3.8, +11.2]* ($n_c = 15$)	+7.4 [+3.8, +11.0]* ($n_c = 15$)	+7.4 [+4.0, +11.1]* ($n_c = 15$)	+7.4 [+4.0, +11.1]* ($n_c = 15$)	+6.3 [+2.4, +10.4]* ($n_c = 15$)
Huasco	+2.6 [+0.2, +4.9]* ($n_c = 10$)	+2.8 [+0.8, +4.8]* ($n_c = 10$)	+2.7 [+0.5, +5.2]* ($n_c = 10$)	+3.1 [+0.1, +6.4]* ($n_c = 10$)	+0.8 [-0.6, +2.2] ($n_c = 5$)
Elqui	+3.6 [+1.4, +5.7]* ($n_c = 25$)	+3.6 [+1.4, +5.8]* ($n_c = 25$)	+3.0 [+0.6, +5.2]* ($n_c = 25$)	+3.2 [+1.1, +5.3]* ($n_c = 25$)	+4.6 [+2.8, +6.4]* ($n_c = 25$)
Limarí	+3.8 [+1.4, +6.1]* ($n_c = 22$)	+3.8 [+1.5, +6.1]* ($n_c = 22$)	+3.6 [+1.2, +6.0]* ($n_c = 22$)	+3.5 [+1.2, +5.6]* ($n_c = 22$)	+2.5 [+0.1, +4.8]* ($n_c = 22$)
Joint (4 targets)	**+4.3**, $p < 0.001$ ($n_c = 72$)	**+4.4**, $p < 0.001$ ($n_c = 72$)	**+4.2**, $p < 0.001$ ($n_c = 72$)	**+4.3**, $p < 0.001$ ($n_c = 72$)	**+3.6**, $p < 0.001$ ($n_c = 67$)

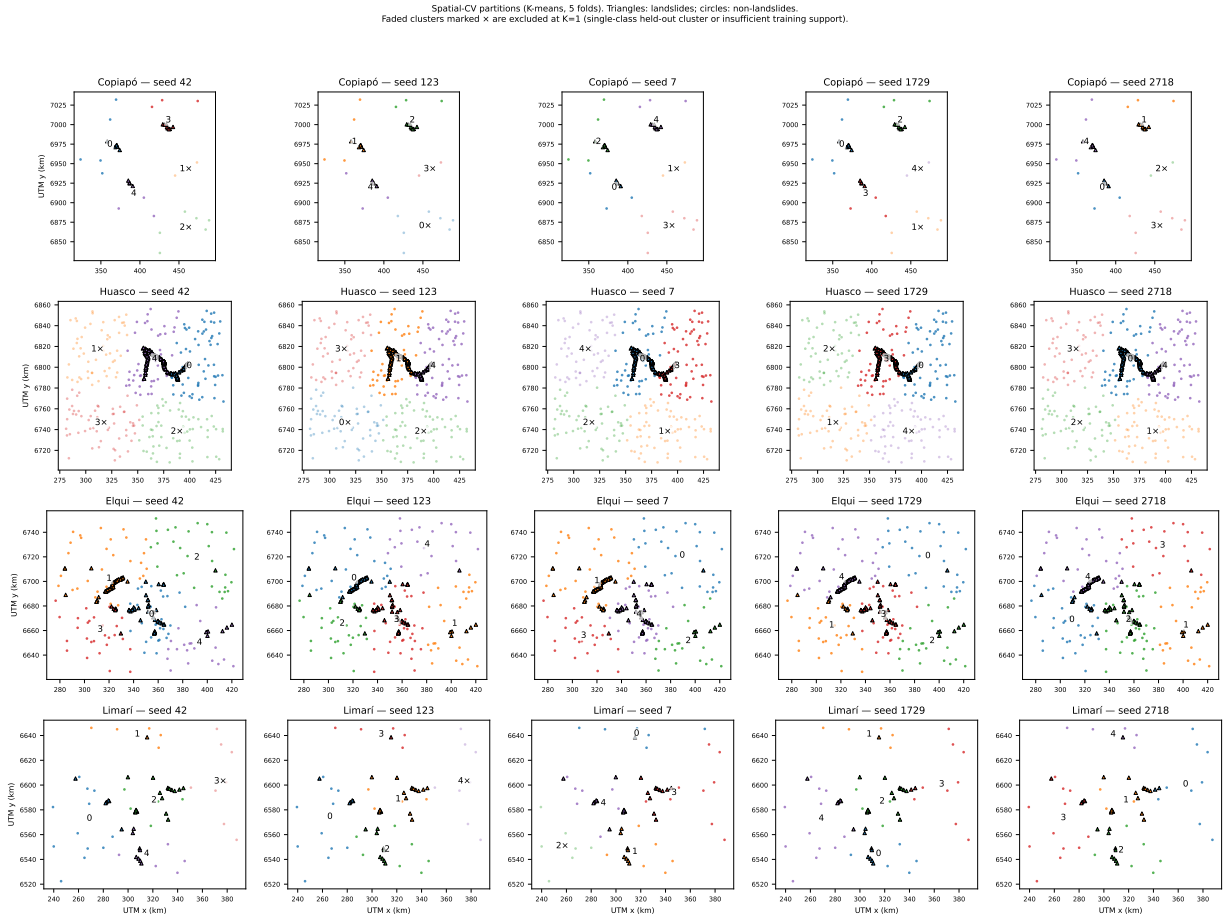


Figure B.8: Spatial-CV partitions (K-means, 5 folds) for each target basin and seed. Triangles: landslide pixels; circles: non-landslide pixels; colours: folds. Faded clusters marked × are excluded at $K = 1$ because the held-out cluster contains a single class; they are the low-relief, event-free parts of Huasco and Limarí (Table A.8). In Huasco the partitions of all seeds are identical (ARI = 1), so its (seed, fold) replicates reduce to two spatial folds.

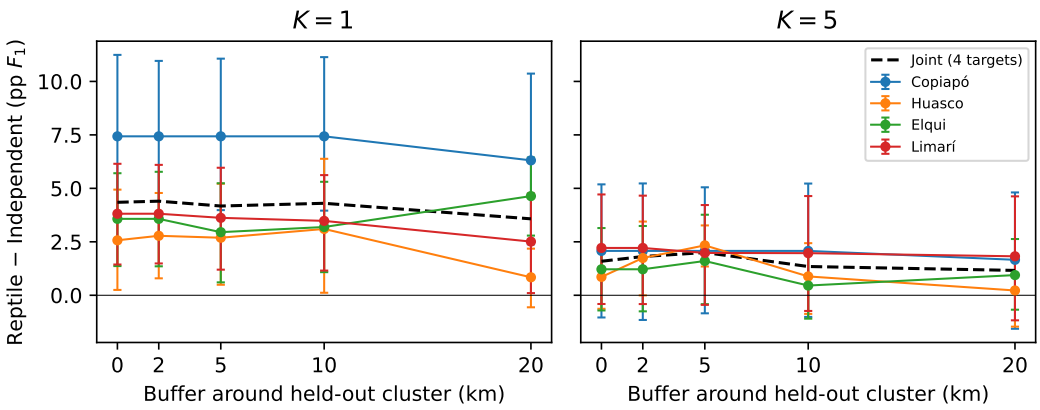


Figure B.9: Buffered spatial cross-validation: Reptile − Independent cluster-level paired difference (pp F_1, 95% hierarchical-bootstrap CI) as a function of the buffer removed around the held-out cluster, at $K = 1$ (left) and $K = 5$ (right); dashed black line: joint cross-basin estimate. Buffers up to 10 km remove $\leq 3.5\%$ of the support pool because K-means folds already separate query and support pixels by ≥ 2 km; the 20 km buffer, comparable to the residual-variogram ranges, removes 14–21% of the support in three basins and attenuates the joint $K = 1$ advantage from +4.3 to +3.6 pp ($p < 0.001$; Table A.10).

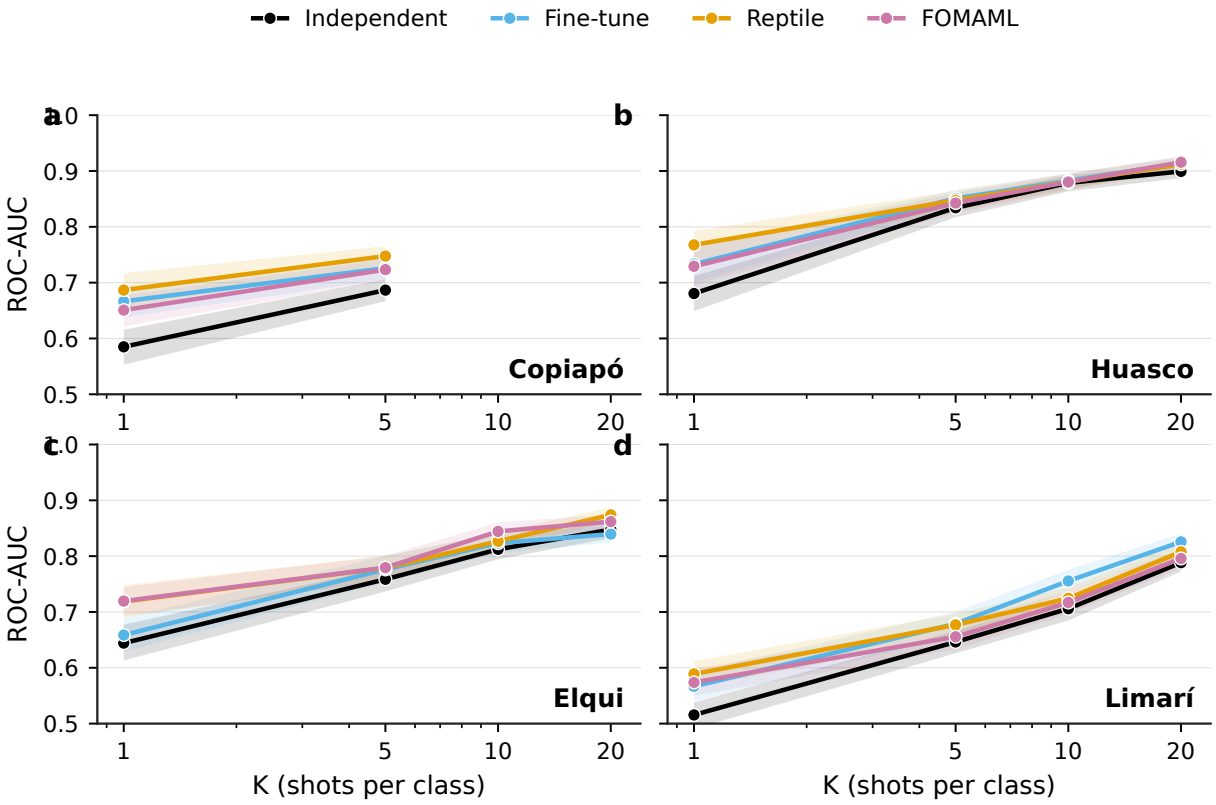


Figure B.10: ROC-AUC versus K under random cross-validation. Pattern consistent with F_1 results in Figure 2.

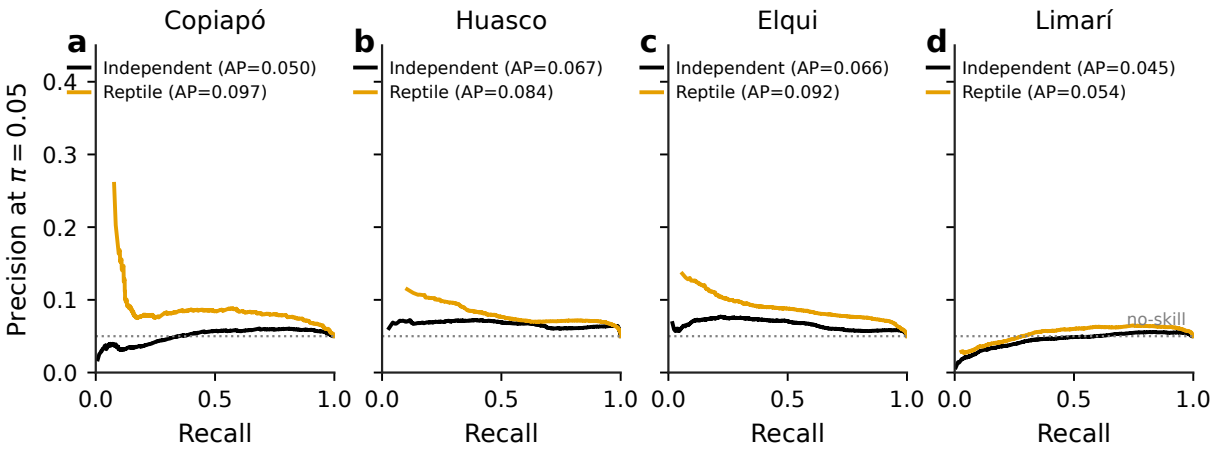


Figure B.11: Precision–recall re-weighted to a realistic landslide prevalence ($\pi = 0.05$) for the pooled $K = 1$ spatial-CV queries (Independent vs. Reptile), per target basin. Curves are obtained from the held-out-query operating characteristics via $\text{precision}(\pi) = \pi\,\text{TPR}/[\pi\,\text{TPR} + (1-\pi)\,\text{FPR}]$; the dotted line marks the no-skill baseline (precision $= \pi$). Absolute precision is low at operational prevalence—most flagged pixels are false positives, a property masked by the fixed-threshold F_1—yet the meta-learned Reptile initialisation dominates Independent in average precision (AP, legend) in all four targets, confirming the $K = 1$ advantage is prevalence-robust (Section 5.7).

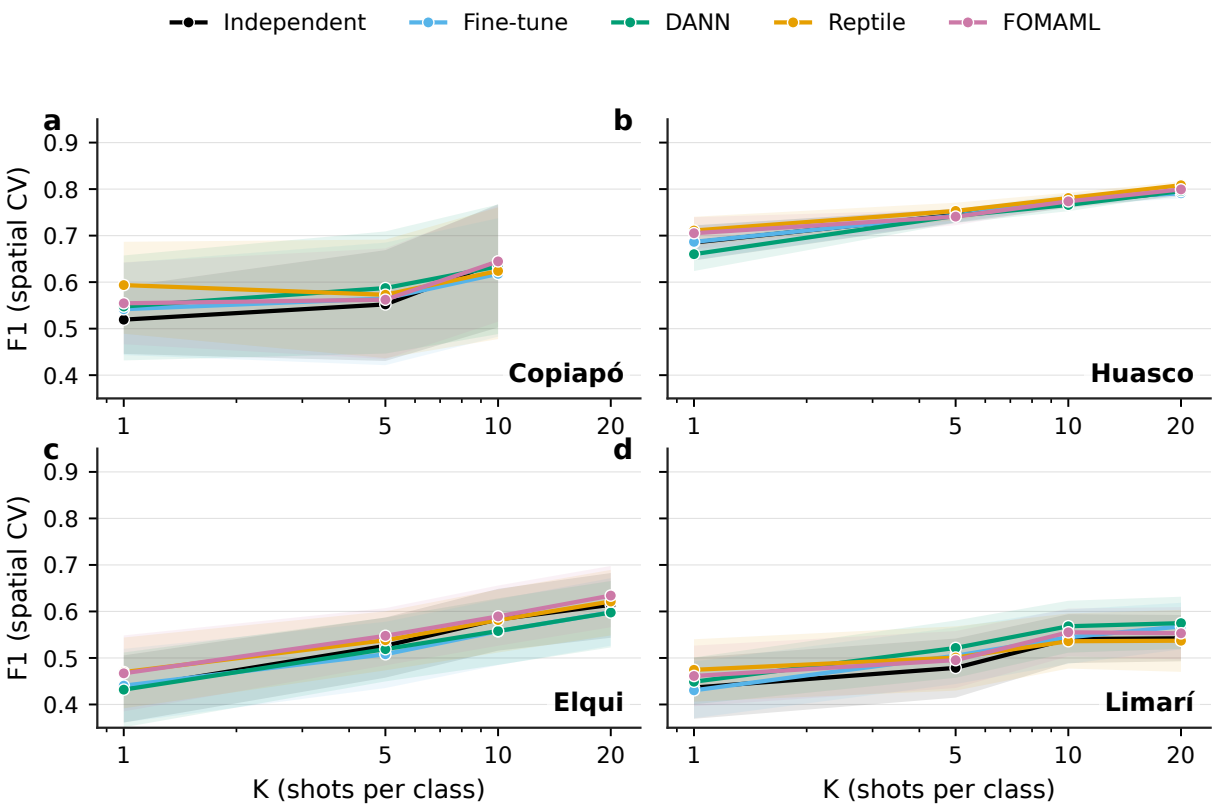


Figure B.12: F_1 versus K under spatial cross-validation, including DANN (source-only features, five seeds). Shaded bands are two-stage hierarchical-bootstrap 95% intervals (clusters, then episodes). Reptile has the highest mean F_1 at $K = 1$ in all four target basins; the per-basin and joint significance of the difference to Independent is given in Table 4.

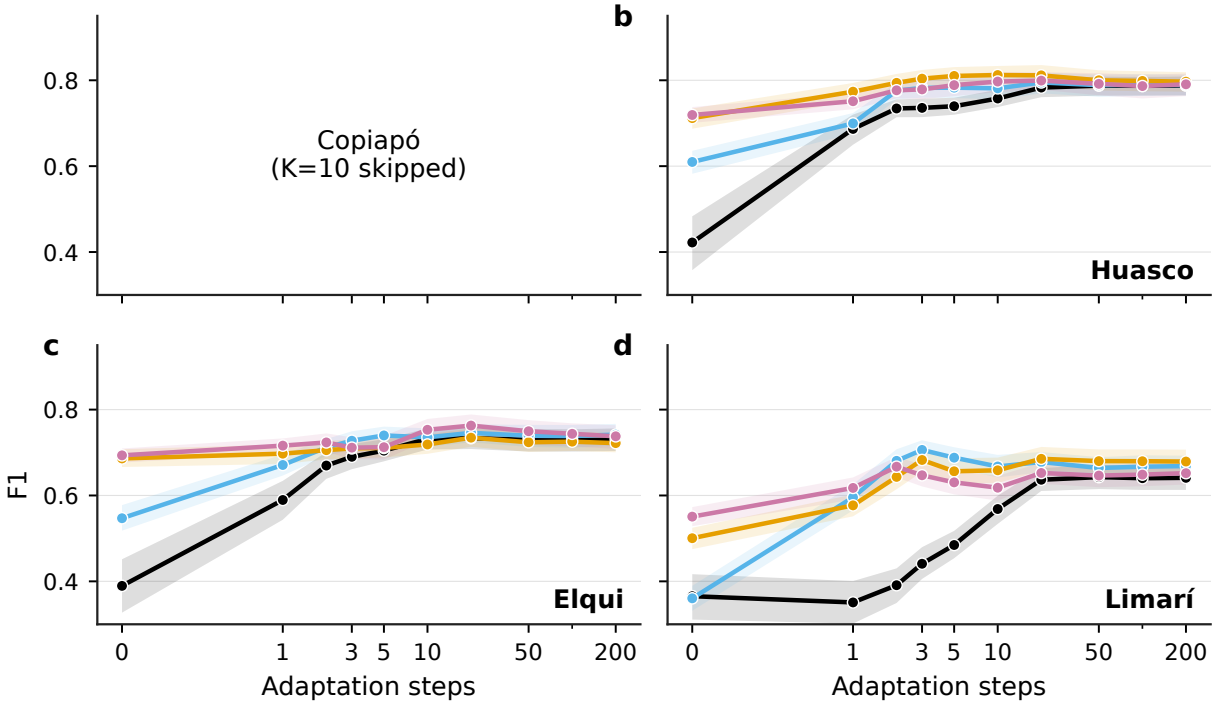


Figure B.13: Adaptation curves at $K = 10$. Same convergence pattern as $K = 5$ (Figure 4).

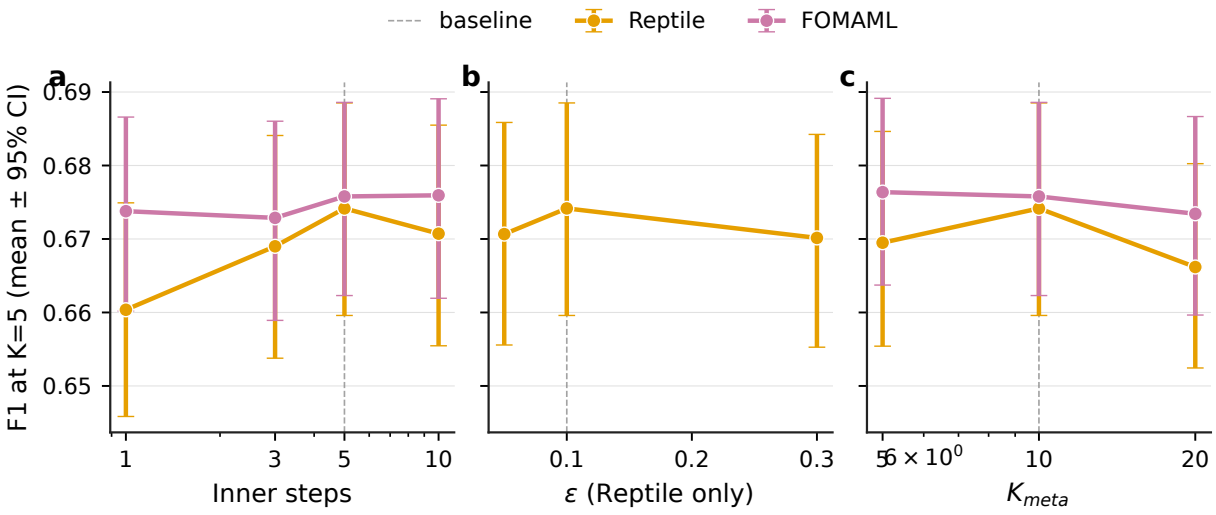


Figure B.14: Hyperparameter sensitivity ablation (random CV, source-only features). (a) Inner gradient steps. (b) Reptile ϵ. (c) Meta-training K. Sensitivity is small (≤ 1.4 pp F_1 across all settings); the default configuration is robust.

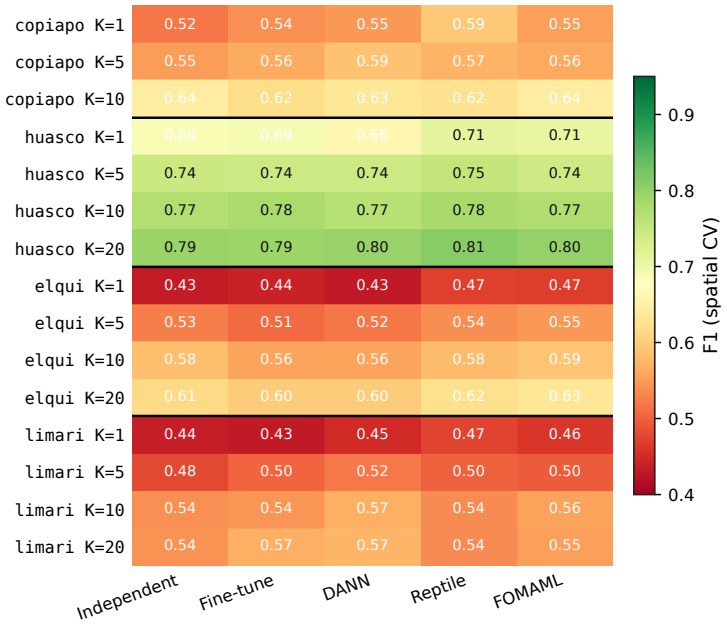


Figure B.15: F_1 (mean) for all (target $\times$ K $\times$ method) combinations under spatial CV.

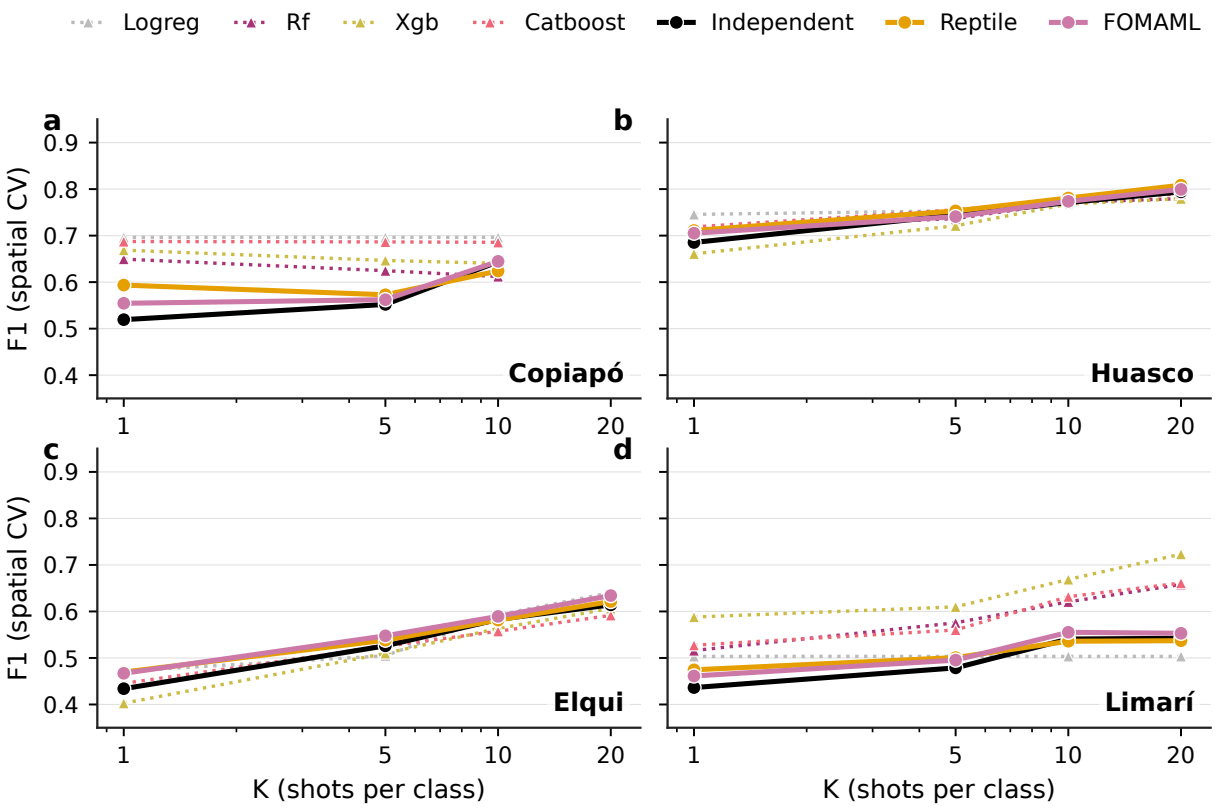


Figure B.16: Classical machine learning vs. meta-learning under spatial CV (source-only features). Logistic regression trained on the pooled source basins plus the K target samples matches or exceeds the meta-learners at $K = 1$ in all four targets; at $K \geq 5$ the meta-learned MLP leads in Elqui, Huasco and Limarí (Section 5.11).

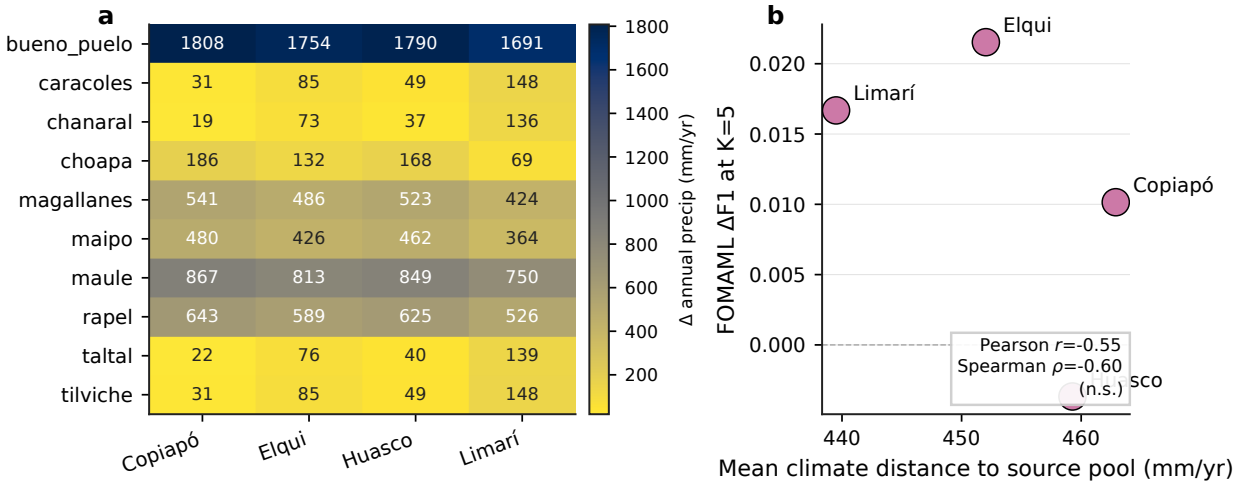


Figure B.17: (a) Climate distance matrix between source and target basins (annual precipitation difference). (b) FOMAML transfer advantage at $K = 5$ vs. mean source-target climate distance; no detectable correlation with $n = 4$ targets (Pearson $r = -0.55$, $p = 0.45$; Spearman $\rho = -0.60$, $p = 0.40$).

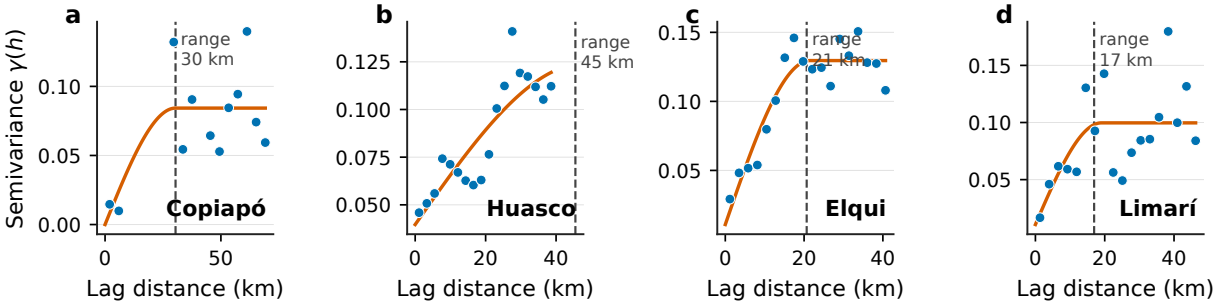


Figure B.18: Empirical variogram of Independent-model residuals for each target basin, with fitted spherical model (red dashed line marks the autocorrelation range). Fitted ranges (source-only features): 30.4 km (Copiapó), 45.5 km (Huasco), 20.7 km (Elqui), 16.9 km (Limarí). At $K_{\text{folds}} = 5$ the median fold-centroid separation exceeds the range in all four basins; the direct query-to-support distances and the buffered validation of Section 5.6 are the primary evidence of spatial separation.

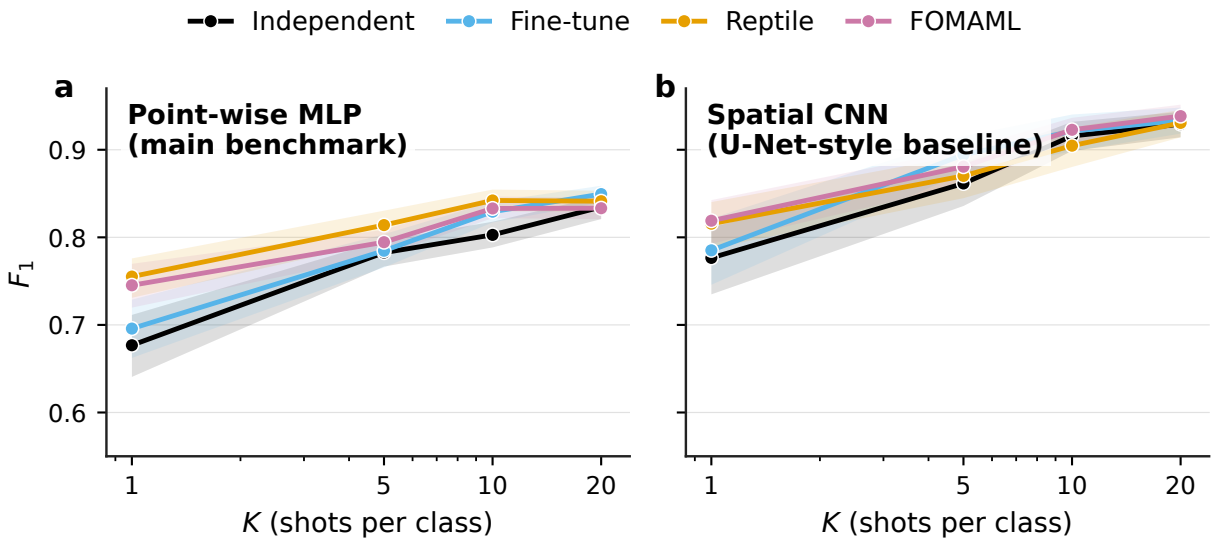


Figure B.19: Spatial CNN (U-Net-style) backbone vs. point-wise MLP on Huasco under the same spatial K-shot CV protocol with 9 source basins for meta-training. The CNN exceeds the MLP by 6–11 percentage points F_1 across all four methods and K values. The meta-learning lift over Independent is smaller in the CNN regime, indicating that spatial context absorbs part of the prediction signal that meta-learning's prior captures in the point-wise setting.

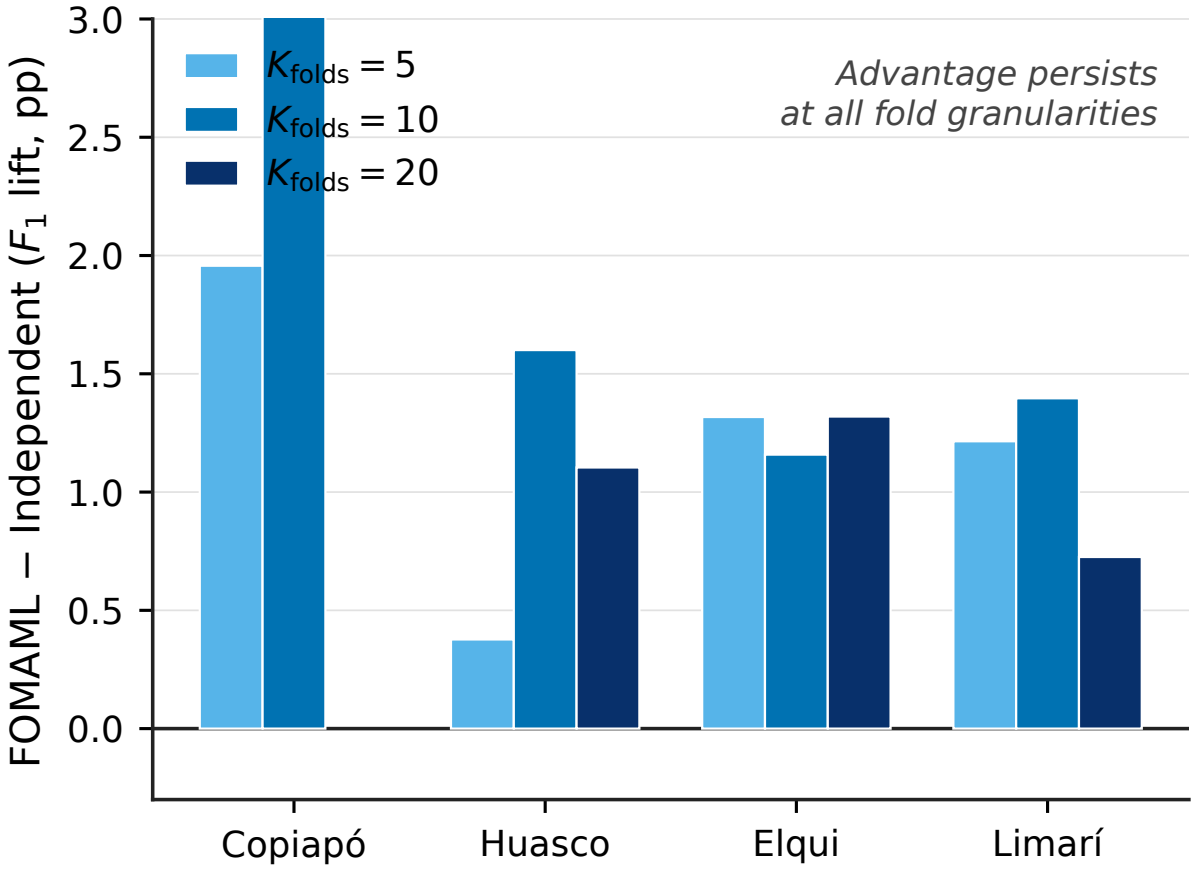


Figure B.20: Reptile and FOMAML − Independent mean F_1 lift as a function of the number of spatial CV folds $K_{\text{folds}} \in \{5, 10, 20\}$, for $K = 1$, 5 and 10 shots (three seeds, 10 episodes per fold). At $K = 1$ the Reptile lift persists at every granularity in Copiapó (+7.9, +5.1 pp), Elqui (+3.5, +2.5, +2.5) and Huasco (+2.4, +2.9, +3.5); in Limarí it is +3.7 pp at 5 folds but −0.5 and +0.7 pp at 10 and 20 folds, where held-out clusters become very small.